

Quantum Simulation of Spin-Dependent Charmonium Spectroscopy with Variational Quantum Algorithms

Master's Thesis

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Contents

1	Introduction	1
1.1	Quantum Computing	1
1.1.1	Why quantum computing?	1
1.1.2	Qubits and quantum states	2
1.1.3	Entanglement	3
1.1.4	Quantum gates and circuits	3
1.1.5	Quantum parallelism and interference	4
1.1.6	Measurement and the Born rule	5
1.1.7	Density matrices and mixed states	5
1.1.8	Quantum simulation	6
1.1.9	The NISQ era	7
1.2	Variational Quantum Methods	7
1.2.1	The variational principle	7
1.2.2	The hybrid quantum–classical loop	8
1.2.3	Ansatz design and practical considerations	9
1.3	The Quarkonium Problem	9
1.3.1	Heavy quark bound states	9
1.3.2	The Cornell potential	9
1.3.3	Spin-dependent interactions	10
1.3.4	Quarkonium as a quantum simulation testbed	11
1.4	Thesis Overview	11
2	Mathematical Framework and Quantum Algorithms	13
2.1	Common Framework	13
2.1.1	The non-relativistic quark model	14
2.1.2	Radial equation and the three channels	14
2.1.3	Harmonic-oscillator basis expansion	15
2.1.4	Matrix elements	16
2.1.5	The three channel Hamiltonian matrices	17
2.1.6	Pauli decomposition of a Hermitian matrix	18
2.2	Gallimore & Liao — Variational Quantum Eigensolver	19

2.2.1	The variational principle	19
2.2.2	Jordan–Wigner encoding	20
2.2.3	Unitary Coupled Cluster ansatz	22
2.2.4	Energy measurement on a quantum processor	23
2.2.5	Classical optimisation	24
2.2.6	Excited states via orthogonalisation	24
2.2.7	Extension to the spin-dependent N=4 case	25
2.3	Woloshyn — Variational Imaginary-Time Evolution	26
2.3.1	Imaginary-time evolution as a ground-state projector	26
2.3.2	Direct (compact) encoding	27
2.3.3	Three-parameter variational ansatz	28
2.3.4	The McLachlan variational principle	29
2.3.5	The quantum metric tensor in practice	31
2.3.6	Energy gradient and the parameter-shift rule	31
2.3.7	Time stepping and regularisation	32
2.3.8	Excited states by penalisation	33
2.3.9	Transition amplitudes	33
2.4	Yi et al. — Trotter–Taylor Imaginary-Time Evolution	36
2.4.1	Strategy: implement the propagator directly	37
2.4.2	Trotter decomposition	37
2.4.3	Taylor expansion of a single Trotter step	37
2.4.4	Linear-combination-of-unitaries (LCU) implementation	38
2.4.5	The full TTITE algorithm	39
2.4.6	Success probability and resource cost	40
2.5	Summary	41
3	Results	44
3.1	Experimental setup	44
3.2	Noiseless benchmark	45
3.2.1	Ground-state energies	45
3.2.2	Ground-state error summary	47
3.2.3	Computational cost	47
3.2.4	Physical interpretation: meson masses	47
3.2.5	Excited states	48
3.3	Transition amplitudes	49
3.4	Noisy benchmark	51
3.5	Summary	54
	Bibliography	55

Chapter 1

Introduction

The simulation of quantum systems lies at the heart of quantum computing’s promise. As Feynman observed in 1982, classical computers face a fundamental barrier when modelling quantum mechanical systems: the Hilbert space of an n -particle system grows exponentially, rendering exact classical computation intractable for even modest system sizes [1]. Quantum computers, by contrast, naturally encode and manipulate quantum states, offering a path toward efficient simulation of physical systems that are otherwise inaccessible [2].

This thesis applies variational quantum algorithms to the spectroscopy of charmonium—bound states of a charm quark and its antiquark—including spin-dependent interactions that produce the experimentally observed mass splittings between states such as the η_c and J/ψ mesons. The quarkonium system is physically rich enough to probe the capabilities of near-term quantum algorithms across multiple quantum number channels and excited states, yet small enough to be tractable with only a few qubits.

The remainder of this chapter introduces the three pillars on which this work rests: the fundamentals of quantum computing (Section 1.1), variational quantum methods (Section 1.2), and the quarkonium problem (Section 1.3).

1.1 Quantum Computing

1.1.1 Why quantum computing?

The classical theory of computation, formalized by Turing and Church in the 1930s, rests on the assumption that information is processed according to classical physics. For decades this abstraction posed no practical limitation. However, the simulation of quantum mechanical systems reveals a fundamental barrier: the state space of an n -particle quantum system is a Hilbert space of dimension 2^n , so that a faithful classical representation requires resources that grow exponentially with system size. A modest system of 50 interacting spin- $\frac{1}{2}$ particles, for instance, demands $2^{50} \approx 10^{15}$ complex amplitudes—

exceeding the memory of any existing classical computer.

In 1982, Feynman proposed a radical alternative: rather than simulating quantum mechanics on a classical machine, one should use a controllable quantum system to simulate another [1]. The exponential scaling that makes classical simulation intractable becomes an *asset*: a quantum processor with n qubits naturally inhabits the same 2^n -dimensional space as the target system. This observation laid the conceptual foundation for quantum computing.

The subsequent development of quantum algorithms by Deutsch, Shor, and Grover [2] demonstrated that quantum computers are not merely faster classical machines, but represent a genuinely different model of computation. The complexity class BQP (bounded-error quantum polynomial time) is believed to be strictly larger than its classical counterpart BPP, meaning there exist problems for which quantum algorithms achieve an exponential advantage that no classical algorithm can match [2]. The source of this advantage lies in two uniquely quantum phenomena: *superposition*, which allows exploration of exponentially many computational paths simultaneously, and *interference*, which enables the selective amplification of correct answers and suppression of incorrect ones.

1.1.2 Qubits and quantum states

The fundamental unit of quantum information is the *qubit*, a two-level quantum system whose pure state is described by a normalized vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1.1)$$

where $\alpha, \beta \in \mathbb{C}$ are *probability amplitudes* and $\{|0\rangle, |1\rangle\}$ is the computational basis (conventionally identified with the eigenstates of the Pauli Z operator). Unlike a classical bit, which is deterministically 0 or 1, a qubit exists in a coherent *superposition* of both basis states.

Geometrically, any single-qubit pure state can be represented as a point on the *Bloch sphere* via the parameterization

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (1.2)$$

with polar angle $\theta \in [0, \pi]$ and azimuthal angle $\phi \in [0, 2\pi)$. The north and south poles correspond to $|0\rangle$ and $|1\rangle$, respectively, while points on the equator represent equal superpositions with varying relative phase.

For a system of n qubits, the state space is the tensor product $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, spanned by 2^n computational basis states $|x_1 x_2 \cdots x_n\rangle$ with $x_i \in \{0, 1\}$. A general pure state is

therefore

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle, \quad \sum_x |c_x|^2 = 1, \quad (1.3)$$

requiring 2^n complex amplitudes to specify. It is this exponential growth of the state space that makes quantum systems both hard to simulate classically and powerful as computational devices.

1.1.3 Entanglement

A multi-qubit state is called *separable* (or a product state) if it can be written as a tensor product of individual qubit states, $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$. States that *cannot* be decomposed in this way are *entangled*. The simplest example is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (1.4)$$

which describes two qubits in a joint superposition: neither qubit has a definite value, yet their measurement outcomes are perfectly correlated. This correlation persists regardless of the spatial separation between the qubits and cannot be explained by any local hidden variable theory, as demonstrated by Bell's theorem [3].

Entanglement is not merely a theoretical curiosity—it is a computational resource. Classical correlations can be shared between bits using shared randomness, but entangled qubits carry correlations that are strictly stronger. In quantum algorithms, entangling gates create superpositions over correlated multi-qubit configurations, enabling the exploration of an exponentially large solution space in a way that has no classical counterpart. Without entanglement, a quantum computer can be efficiently simulated classically [2]; it is entanglement that unlocks the full power of quantum computation.

1.1.4 Quantum gates and circuits

The time evolution of a closed quantum system is governed by a unitary operator. Quantum computation exploits this by decomposing the desired evolution into a sequence of elementary *quantum gates*: unitary transformations acting on one or two qubits at a time.

The three Pauli operators form a basis for single-qubit operations:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.5)$$

Together with the identity I , they satisfy $\sigma_i^2 = I$, anticommute pairwise ($\sigma_i \sigma_j = -\sigma_j \sigma_i$ for $i \neq j$), and form a basis for the space of 2×2 Hermitian matrices. Any 2×2 Hermitian operator—and hence any single-qubit observable—can be written as $A = a_0 I + a_1 X + a_2 Y + a_3 Z$ with $a_i \in \mathbb{R}$.

The Pauli operators generate continuous families of rotation gates on the Bloch sphere:

$$R_\mu(\theta) = e^{-i\theta\mu/2} = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}\mu, \quad \mu \in \{X, Y, Z\}. \quad (1.6)$$

The Hadamard gate $H = (X + Z)/\sqrt{2}$ maps $|0\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$, creating an equal superposition from a computational basis state.

Two-qubit entangling gates introduce correlations between qubits. The controlled-NOT (CNOT) gate flips the target qubit conditioned on the control qubit being in state $|1\rangle$:

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1.7)$$

The set $\{R_X, R_Y, R_Z, \text{CNOT}\}$ constitutes a *universal gate set*: any unitary on n qubits can be approximated to arbitrary precision by a finite sequence of these gates [2].

A quantum algorithm is specified as a *quantum circuit*: a sequence of gates applied to a register of qubits initialized in a known state, followed by measurement. Figure 1.1 illustrates a minimal circuit that prepares the Bell state of Eq. (1.4) using only a Hadamard gate and a CNOT.

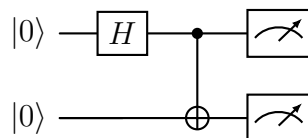


Figure 1.1: Quantum circuit preparing the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The Hadamard gate creates a superposition on the first qubit; the CNOT entangles both qubits. Measurement yields $|00\rangle$ or $|11\rangle$, each with probability $1/2$.

1.1.5 Quantum parallelism and interference

The mechanism behind quantum computational advantage is often described loosely as “computing on all inputs at once.” While evocative, this picture is incomplete: the key ingredient is not parallelism alone, but the ability to make the parallel branches *interfere*.

Consider a unitary U_f that evaluates a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. Applied to an equal superposition over all 2^n inputs, it produces

$$U_f \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |f(x)\rangle. \quad (1.8)$$

All function values are encoded in a single quantum state. However, measuring this state yields a *single* random pair $(x, f(x))$ —no better than classical evaluation. The power of

quantum algorithms lies in subsequent operations that exploit *interference*: unitary gates are designed so that amplitudes corresponding to desired outputs add constructively, while those corresponding to undesired outputs cancel destructively. The final measurement then yields the correct answer with high probability.

This interplay of superposition, entanglement, and interference is the mathematical core of quantum speedups. Superposition provides exponential parallelism; entanglement enables correlations across qubits that have no classical equivalent; and interference channels the computation toward the correct outcome.

1.1.6 Measurement and the Born rule

While unitary evolution is deterministic, extracting information from a quantum system is inherently probabilistic. A *projective measurement* in the computational basis is described by the set of projectors $\{\Pi_x = |x\rangle\langle x|\}_{x \in \{0,1\}^n}$. Given a state $|\Psi\rangle = \sum_x c_x |x\rangle$, the *Born rule* prescribes the outcome probabilities:

$$p(x) = |\langle x|\Psi\rangle|^2 = |c_x|^2. \quad (1.9)$$

Upon obtaining outcome x , the state collapses to $|x\rangle$, and any information encoded in the remaining amplitudes is irretrievably lost.

This has a direct practical consequence: a single measurement of n qubits yields only n classical bits, despite the state containing 2^n amplitudes. To estimate a continuous quantity—such as an energy expectation value—the measurement must be repeated many times (N_{shots} repetitions), and the result is subject to statistical uncertainty scaling as $1/\sqrt{N_{\text{shots}}}$.

More generally, to measure an observable O (a Hermitian operator), its expectation value in state $|\Psi\rangle$ is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle = \sum_i \lambda_i p_i, \quad (1.10)$$

where λ_i are the eigenvalues of O and p_i the probability of obtaining the corresponding eigenstate. When O is decomposed as a sum of Pauli operators, each Pauli term can be measured independently by rotating to the appropriate eigenbasis before measurement in the computational basis.

1.1.7 Density matrices and mixed states

The pure state formalism of Eq. (1.1) describes isolated, perfectly controlled quantum systems. Real quantum processors, however, interact with their environment, leading to *decoherence* and *mixed states* that require a more general description.

The *density matrix* (or density operator) of a pure state $|\psi\rangle$ is

$$\rho = |\psi\rangle\langle\psi|. \quad (1.11)$$

A *mixed state*—representing classical uncertainty about which pure state the system is in—is described by a convex combination

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \quad (1.12)$$

The density matrix is Hermitian, positive semidefinite, and has unit trace. A state is pure if and only if $\text{Tr}(\rho^2) = 1$; for mixed states, $\text{Tr}(\rho^2) < 1$, quantifying the degree of decoherence.

In the density matrix formalism, unitary evolution becomes $\rho \mapsto U\rho U^\dagger$, and expectation values are computed as $\langle O \rangle = \text{Tr}(O\rho)$. Noise processes—such as the depolarizing channel relevant to this thesis—are described by completely positive trace-preserving (CPTP) maps on ρ . For example, the single-qubit depolarizing channel with error rate p acts as

$$\mathcal{E}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z), \quad (1.13)$$

which with probability p replaces the state with a random Pauli error. This formalism will be central to the analysis of noise effects in Chapter 1 and subsequent chapters.

1.1.8 Quantum simulation

Among the many proposed applications of quantum computing—including cryptanalysis, optimization, and machine learning—the simulation of quantum systems stands out as the most natural and arguably the most promising near-term application [1, 4]. The reason is both conceptual and practical: a quantum computer *is* a quantum system, so using it to simulate another quantum system leverages its native capabilities rather than working against them.

The essential mapping is straightforward. A physical system with a 2^n -dimensional Hilbert space is encoded on n qubits. The system’s Hamiltonian H is decomposed into a sum of Pauli operators acting on these qubits:

$$H = \sum_{i=1}^M c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n}, \quad (1.14)$$

where each P_i is a tensor product of single-qubit Pauli matrices (a *Pauli string*) and $c_i \in \mathbb{R}$ are real coefficients. This decomposition is always possible since the 4^n Pauli strings form a complete basis for $2^n \times 2^n$ Hermitian matrices.

Ground state energies, excited state spectra, and dynamical properties can then be

extracted using quantum algorithms that prepare and measure appropriate states on the quantum register. For the small systems considered in this thesis, the encoding requires only 2–3 qubits, but the same framework scales to larger systems where classical methods become intractable.

Quantum simulation of atomic, molecular, and nuclear systems has seen rapid progress in recent years, with demonstrations ranging from small molecules to lattice gauge theories [5]. The quarkonium system studied in this thesis belongs to this broader programme, representing a step toward quantum simulation of QCD bound states.

1.1.9 The NISQ era

Current quantum hardware operates in the *Noisy Intermediate-Scale Quantum* (NISQ) regime [4], characterized by processors with tens to hundreds of qubits subject to significant imperfections:

- **Gate errors** on the order of 10^{-3} – 10^{-2} per operation, described by noise channels such as the depolarizing channel of Eq. (1.13);
- **Limited connectivity** between qubits, requiring additional SWAP gates that increase circuit depth;
- **Short coherence times** that bound the number of operations executable before decoherence degrades the quantum state.

These constraints preclude the deep circuits required by fault-tolerant algorithms such as quantum phase estimation, which need thousands of controlled operations and ancilla qubits for error correction.

The NISQ era has therefore driven the development of algorithms specifically designed for shallow circuits and resilient to moderate noise levels. The most successful paradigm to date is the *variational* approach, in which a quantum processor prepares parameterized trial states while a classical computer optimizes the parameters—a hybrid strategy discussed in the next section.

1.2 Variational Quantum Methods

1.2.1 The variational principle

Variational methods offer a framework for extracting useful results from NISQ hardware. The central idea rests on the *Rayleigh–Ritz variational principle*: for a Hamiltonian H with ground state energy E_0 ,

$$E_0 \leq \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle \equiv E(\boldsymbol{\theta}) \quad (1.15)$$

for any normalized trial state $|\psi(\boldsymbol{\theta})\rangle$ parameterized by classical variables $\boldsymbol{\theta} \in \mathbb{R}^p$. Minimizing $E(\boldsymbol{\theta})$ over the parameter space yields an upper bound on E_0 , with equality when $|\psi(\boldsymbol{\theta})\rangle$ includes the true ground state in its span.

1.2.2 The hybrid quantum–classical loop

In the quantum computing implementation [5, 6], the trial state is prepared by a *parameterized quantum circuit* (also called an *ansatz*) applied to an initial reference state:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle^{\otimes n}, \quad (1.16)$$

where $U(\boldsymbol{\theta})$ is a product of parameterized and fixed gates. The energy expectation value is then estimated on the quantum processor. For a Hamiltonian decomposed as a weighted sum of Pauli strings,

$$H = \sum_{i=1}^M c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n}, \quad (1.17)$$

each expectation value $\langle P_i \rangle$ is estimated independently through repeated state preparation and measurement in the appropriate Pauli basis. The total energy,

$$E(\boldsymbol{\theta}) = \sum_{i=1}^M c_i \langle P_i \rangle_{\boldsymbol{\theta}}, \quad (1.18)$$

is passed to a *classical optimizer* that updates $\boldsymbol{\theta}$ to minimize E . This loop repeats until convergence, as illustrated in Fig. 1.2.

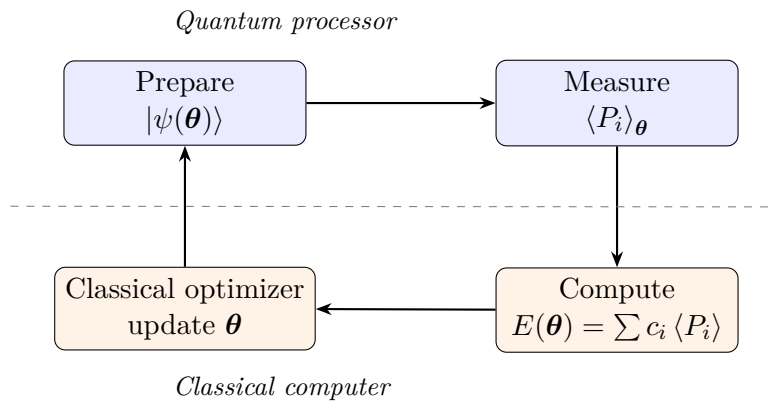


Figure 1.2: Schematic of the hybrid quantum–classical variational loop. The quantum processor prepares a parameterized state and estimates Pauli expectation values; the classical computer evaluates the energy and updates circuit parameters. The loop iterates until convergence.

1.2.3 Ansatz design and practical considerations

The choice of ansatz circuit is problem-dependent and critical to performance. *Hardware-efficient* ansätze use gates native to the target device, minimizing circuit depth at the cost of physical motivation. *Physically-motivated* ansätze, such as those inspired by unitary coupled cluster theory from quantum chemistry, encode domain knowledge into the circuit structure and may converge faster toward the physical ground state.

The hybrid approach offers several advantages for near-term hardware. The quantum processor handles the exponentially hard task of preparing and measuring many-body quantum states, while the classical optimizer navigates the energy landscape using well-established numerical methods. Circuit depths remain manageable, and variational methods exhibit a degree of inherent noise resilience: small perturbations in the energy surface due to hardware errors are partially absorbed by the optimization [6].

Beyond the ground state, variational frameworks extend naturally to excited states through orthogonality constraints and penalty methods, and to quantum dynamics through variational formulations of time evolution. These extensions broaden the scope of problems addressable on near-term quantum hardware.

1.3 The Quarkonium Problem

1.3.1 Heavy quark bound states

Quantum chromodynamics (QCD), the gauge theory of the strong interaction, governs the binding of quarks and gluons into hadrons. Among the simplest hadronic systems are *quarkonia*: bound states of a heavy quark and its antiquark ($q\bar{q}$). Charmonium ($c\bar{c}$) systems have served as testing grounds for QCD since the simultaneous discovery of the J/ψ meson in 1974 at BNL [7] and SLAC [8]—an event known as the *November Revolution* in particle physics.

The large mass of the charm quark ($m_c \approx 1.27 \text{ GeV}/c^2$) implies that the quark–antiquark pair moves non-relativistically inside the bound state, with typical velocities $v/c \sim 0.3$. This justifies a potential model approach [9] in which the dynamics are governed by the non-relativistic Schrödinger equation with an effective interquark potential.

1.3.2 The Cornell potential

The *Cornell potential* [9],

$$V(r) = -\frac{\kappa}{r} + \sigma r, \quad (1.19)$$

provides a successful phenomenological description of the $q\bar{q}$ interaction. It captures the two essential features of the strong force:

- **Short-range Coulomb attraction** ($-\kappa/r$), arising from one-gluon exchange between the color charges, analogous to the electromagnetic Coulomb potential but with QCD coupling;
- **Long-range linear confinement** (σr), a consequence of the non-Abelian nature of QCD, modelled by a chromoelectric flux tube (string) with tension σ connecting the quark and antiquark.

Typical parameter values for charmonium are $\kappa \approx 0.52$ and $\sigma \approx 0.183 \text{ GeV}^2$ [9]. Figure 1.3 shows the characteristic shape of this potential.

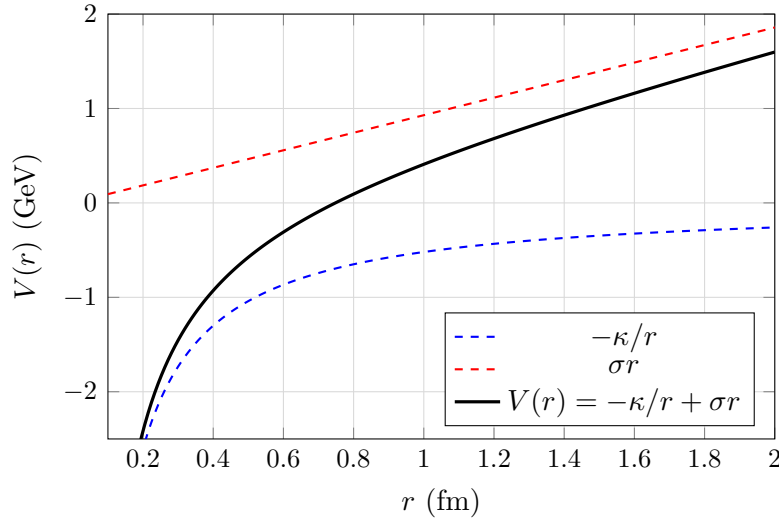


Figure 1.3: The Cornell potential as a function of interquark separation r . The dashed curves show the Coulomb ($-\kappa/r$, blue) and linear confinement (σr , red) components; the solid curve is their sum. At short distances the potential is dominated by one-gluon exchange; at large distances the linearly rising string tension ensures quark confinement.

1.3.3 Spin-dependent interactions

The spin-independent Cornell potential alone does not account for the full charmonium spectrum. The mass splitting between states with different spin quantum numbers—for instance, between the η_c (1S_0 , spin singlet) and the J/ψ (3S_1 , spin triplet)—requires spin-dependent corrections. At leading order, the dominant contribution for S -wave states is the hyperfine contact interaction [10],

$$V_{\text{hyp}}(r) = \frac{32\pi \alpha_s}{9 m_c^2} \delta^{(3)}(\mathbf{r}) \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \quad (1.20)$$

where α_s is the strong coupling constant and $\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$ is the spin–spin dot product. The eigenvalues of the latter operator distinguish singlet and triplet configurations:

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S = 0), \\ +\frac{1}{4} & \text{spin triplet } (S = 1). \end{cases} \quad (1.21)$$

Quarkonium states are classified in spectroscopic notation $^{2S+1}L_J$, where S is the total spin, L the orbital angular momentum ($S, P, D, \dots$ for $L = 0, 1, 2, \dots$), and J the total angular momentum. The low-lying charmonium states relevant to this work are:

- 1S_0 : spin singlet, S -wave $\rightarrow \eta_c$ meson ($M \approx 2984$ MeV),
- 3S_1 : spin triplet, S -wave $\rightarrow J/\psi$ meson ($M \approx 3097$ MeV),
- 1P_1 : spin singlet, P -wave $\rightarrow h_c$ meson ($M \approx 3526$ MeV).

1.3.4 Quarkonium as a quantum simulation testbed

The full quarkonium Hamiltonian, including kinetic energy and both spin-independent and spin-dependent potentials, reads

$$H = -\frac{1}{2\mu}\nabla^2 + V(r) + V_{\text{hyp}}(r), \quad (1.22)$$

where $\mu = m_c/2$ is the reduced mass. Expanding the radial wavefunction in a truncated basis of N quantum harmonic oscillator (QHO) eigenstates yields a finite-dimensional matrix eigenvalue problem of size $N \times N$, whose entries are determined by the matrix elements of the kinetic and potential operators in the QHO basis.

This matrix can then be decomposed into a sum of Pauli operators (Eq. (1.17)) and encoded on a quantum register of $\lceil \log_2 N \rceil$ qubits, making it directly amenable to variational quantum algorithms. With $N = 3$ – 4 basis states, the problem requires only 2–3 qubits, placing it firmly within reach of current NISQ devices while retaining the essential physics: multiple spectroscopic channels, spin-dependent splittings, and a hierarchy of excited states.

This combination of physical richness and computational tractability makes quarkonium an ideal testbed for benchmarking quantum simulation algorithms against exact solutions and experimental data.

1.4 Thesis Overview

This thesis investigates the application of variational quantum algorithms to the charmonium spectroscopy problem described above. We implement and compare several al-

algorithmic approaches for determining both ground and excited state energies across the 1S_0 , 3S_1 , and 1P_1 channels, assess their performance under realistic noise conditions, and evaluate error mitigation strategies. The goal is twofold: to demonstrate the feasibility of quantum simulation for hadronic physics on near-term hardware, and to provide a systematic comparison of methods that may inform future applications to more complex systems.

The thesis is organized as follows:

- **Chapter 2** develops the theoretical framework, including the Hamiltonian construction in the harmonic oscillator basis and the mapping to qubit operators.
- **Chapter 3** presents the quantum algorithms employed, covering both eigenvalue-based and time-evolution-based variational approaches.
- **Chapter 4** details the numerical results: energy spectra, convergence properties, and cross-method comparisons.
- **Chapter 5** addresses the effects of hardware noise and discusses error mitigation techniques.
- **Chapter 6** concludes with a summary of findings and an outlook on future directions.

Chapter 2

Mathematical Framework and Quantum Algorithms

This chapter develops the mathematical machinery used by three quantum algorithms for charmonium spectroscopy that we reproduce in this thesis: the Variational Quantum Eigensolver of Gallimore and Liao [11], the Variational Imaginary-Time Evolution algorithm applied to charmonium by Woloshyn [12], and the Trotter–Taylor Imaginary-Time Evolution scheme of Yi et al. [13]. Each method takes a different approach to estimating the eigenvalues of a Hamiltonian matrix on a quantum processor, and each makes a different choice of qubit encoding for that matrix. We therefore organise the chapter so that the material common to the three methods is presented up front (Section 2.1); each method then receives a dedicated section that develops it in self-contained form, including the specific encoding it uses, its ansatz (if any), its excited-state strategy, and any auxiliary topics treated by the source paper. The Gallimore–Liao section (Section 2.2) presents VQE together with the Jordan–Wigner encoding and an explicit extension of their construction to the spin-dependent $N = 4$ case. The Woloshyn section (Section 2.3) covers VQITE in the direct encoding and includes the M1 and E1 transition amplitudes that Woloshyn also derives. The Yi et al. section (Section 2.4) reproduces TTITE, applied to the same channel Hamiltonians used by the previous two methods. A side-by-side summary closes the chapter (Section 2.5).

2.1 Common Framework

Before introducing the three algorithms, we collect the material they all build on: the charmonium Hamiltonian in a finite harmonic-oscillator basis, the three 4×4 channel matrices, and the general prescription for expressing a Hermitian matrix as a sum of Pauli strings on a quantum register.

2.1.1 The non-relativistic quark model

A charmonium meson is a bound state of a charm quark and its antiquark [9, 10]. Because the charm mass $m_c \simeq 1.48 \text{ GeV}/c^2$ is large compared with the binding energy, the internal motion is well described non-relativistically: the relative coordinate $\mathbf{r} = \mathbf{r}_c - \mathbf{r}_{\bar{c}}$ satisfies the two-body Schrödinger equation

$$H \psi(\mathbf{r}) = \left[-\frac{\nabla^2}{2\mu} + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad (2.1)$$

with reduced mass $\mu = m_c/2$. The interaction $V(r)$ has two pieces. The first is the spin-independent Cornell potential

$$V_0(r) = -\frac{a}{r} + b r, \quad a = \frac{4\alpha_s}{3}, \quad (2.2)$$

where $-a/r$ is the colour-Coulomb attraction from one-gluon exchange and $b r$ is the linearly rising flux-tube term that confines the quarks [9]. The second piece is the leading spin-dependent correction. From the non-relativistic reduction of the Breit interaction, the dominant term at one-gluon-exchange order is the contact (hyperfine) interaction [10, 14]

$$V_s(r) = \frac{32\pi \alpha_s}{9 m_c^2} \delta_\sigma(\mathbf{r}) \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \quad (2.3)$$

where we have replaced the singular Dirac delta with a normalised Gaussian of width σ ,

$$\delta_\sigma(\mathbf{r}) = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 e^{-\sigma^2 r^2}, \quad (2.4)$$

to regularise the contact term inside our finite basis. The dot product of the two quark spin operators is fixed by the total spin $\mathbf{S} = \mathbf{S}_c + \mathbf{S}_{\bar{c}}$ through the identity $\mathbf{S}^2 = \mathbf{S}_c^2 + \mathbf{S}_{\bar{c}}^2 + 2 \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$, giving

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S = 0), \\ +\frac{1}{4} & \text{spin triplet } (S = 1). \end{cases} \quad (2.5)$$

The phenomenological parameters α_s , b , m_c , and σ used throughout this work are those of Barnes, Godfrey, and Swanson [14] and are listed in Table 2.1.

2.1.2 Radial equation and the three channels

Because $V_0(r)$ depends only on $r = |\mathbf{r}|$ and the spin operator in Eq. (2.3) commutes with the orbital angular momentum, eigenstates of H can be classified by the spectroscopic quantum numbers $^{2S+1}L_J$. Writing the wavefunction as

$$\psi(\mathbf{r}) = R_{n\ell}(r) Y_{\ell m}(\hat{\mathbf{r}}) \chi_{SM_S}, \quad (2.6)$$

Table 2.1: Cornell-potential parameters used in this work, following the charmonium fit of Barnes, Godfrey, and Swanson [14]. The conversion $\hbar c = 197.326 \text{ MeV}\cdot\text{fm}$ relates inverse-femtometres to MeV. The Coulomb coefficient is $a = 4\alpha_s/3$.

Parameter	Symbol	Value
Strong coupling	α_s	0.5461
String tension	b	0.1425 GeV ²
Charm-quark mass	m_c	1.4794 GeV
Gaussian width	σ	1.0946 GeV

and defining the reduced radial function $u_{n\ell}(r) = rR_{n\ell}(r)$, the Schrödinger equation reduces to the one-dimensional eigenvalue problem

$$\left[-\frac{1}{2\mu} \frac{d^2}{dr^2} + V_0(r) + \frac{\ell(\ell+1)}{2\mu r^2} + \langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle \frac{32\pi\alpha_s}{9m_c^2} \delta_\sigma(r) \right] u_{n\ell}(r) = E_{n\ell} u_{n\ell}(r), \quad (2.7)$$

where the centrifugal barrier $\ell(\ell+1)/(2\mu r^2)$ is the standard remnant of the angular Laplacian, and the spin expectation value takes the discrete values of Eq. (2.5).

In this work we focus on the three low-lying channels that have served as benchmarks for quantum simulations of quarkonium [12]:

- 1S_0 (associated with the η_c meson): $\ell = 0$, $S = 0$. The centrifugal barrier vanishes and the contact term contributes with $\langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle = -3/4$.
- 3S_1 (associated with the J/ψ meson): $\ell = 0$, $S = 1$. As for 1S_0 but with $\langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle = +1/4$, producing the celebrated η_c - J/ψ hyperfine splitting.
- 1P_1 (associated with the h_c meson): $\ell = 1$, $S = 0$. The contact term contributes only at the origin, and because the $\ell = 1$ wavefunction vanishes there the spin-dependent term is numerically negligible for P -waves; the centrifugal barrier, however, shifts the energies up substantially.

Each channel thus generates a different one-dimensional radial Hamiltonian, which we now discretise.

2.1.3 Harmonic-oscillator basis expansion

The radial functions $u_{n\ell}(r)$ admit a convenient expansion in the eigenstates of a three-dimensional isotropic harmonic oscillator (HO),

$$\phi_{n\ell}(r) = N_{n\ell} r^{\ell+1} \exp\left(-\frac{r^2}{2b^2}\right) L_n^{\ell+1/2}\left(\frac{r^2}{b^2}\right), \quad (2.8)$$

with normalisation $N_{n\ell} = \sqrt{2n! b^{-(2\ell+3)} / \Gamma(n + \ell + 3/2)}$ and L_n^α the associated Laguerre polynomials. The length scale $b = (\mu\omega)^{-1/2}$ is determined by the oscillator frequency ω ,

which is chosen variationally so that the ground state of the basis already captures most of the physical ground-state wavefunction [11].

We truncate the basis to the first $N = 4$ states for each channel,

$$u_{n\ell}(r) \approx \sum_{k=0}^{N-1} c_k^{(n)} \phi_{k\ell}(r), \quad N = 4, \quad (2.9)$$

which turns Eq. (2.7) into the finite-dimensional eigenvalue problem

$$\sum_{k=0}^{N-1} H_{mk}^{(\ell,S)} c_k^{(n)} = E_n c_m^{(n)}, \quad (2.10)$$

where the channel-dependent matrix elements are

$$H_{mk}^{(\ell,S)} = \langle \phi_{m\ell} | T | \phi_{k\ell} \rangle + \langle \phi_{m\ell} | V_0 | \phi_{k\ell} \rangle + \langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle \frac{32\pi\alpha_s}{9m_c^2} \langle \phi_{m\ell} | \delta_\sigma | \phi_{k\ell} \rangle. \quad (2.11)$$

Here T denotes the kinetic term plus, for $\ell > 0$, the centrifugal barrier.

Remark (Hylleraas–Undheim–MacDonald theorem). The truncation in Eq. (2.9) introduces a controlled approximation: the eigenvalues of the truncated matrix $H^{(\ell,S)} \in \mathbb{R}^{N \times N}$ are guaranteed to be upper bounds on the corresponding exact eigenvalues, and they are monotonically non-increasing in N [15]. Increasing N therefore always improves the variational estimate; $N = 4$ provides a useful balance between physical accuracy and the qubit cost of the encodings used by the three methods that follow.

2.1.4 Matrix elements

The matrix elements in Eq. (2.11) can be evaluated in closed form using standard identities for the Laguerre polynomials and the hypergeometric function. We collect the key results here; complete derivations follow the methodology of Gallimore and Liao [11].

Kinetic energy. The kinetic operator, expressed through the ladder operators of the oscillator, is tridiagonal in the HO basis. For $\ell = 0$ one finds

$$\langle m | T | k \rangle = \frac{\omega}{2} \left[\left(2k + \frac{3}{2} \right) \delta_{mk} - \sqrt{k(k + \frac{1}{2})} \delta_{m,k-1} - \sqrt{(k+1)(k + \frac{3}{2})} \delta_{m,k+1} \right]. \quad (2.12)$$

For $\ell > 0$ the centrifugal term is absorbed into a shifted diagonal, $2k + \ell + 3/2$, with the off-diagonal couplings inheriting the same square-root structure with the index $\ell + 1/2$ in place of $1/2$.

Coulomb term. The matrix element of $1/r$ can be expressed in terms of a generalised hypergeometric function ${}_3F_2$ [11]:

$$\langle m|r^{-1}|k\rangle = \frac{(-1)^{m+k} 4b^{-1}}{\pi(1+2k)} \sqrt{\frac{\Gamma(m+\frac{3}{2})\Gamma(k+\frac{3}{2})}{m!k!}} {}_3F_2\left(\frac{1}{2}, 1, -m; \frac{3}{2}, \frac{1}{2}-k; 1\right). \quad (2.13)$$

Linear-confinement term. The matrix element of r involves the Gauss hypergeometric ${}_2F_1$:

$$\langle m|r|k\rangle = \frac{(-1)^{m+k} 4b}{\pi(1-4k^2)} \sqrt{\frac{\Gamma(m+\frac{3}{2})\Gamma(k+\frac{3}{2})}{m!k!}} {}_2F_1\left(2, -m; \frac{3}{2}-k; 1\right). \quad (2.14)$$

Contact term. For the regularised contact interaction with the Gaussian of Eq. (2.4), the matrix element reduces to a single Gaussian integral over a polynomial in r^2 , which is again closed-form:

$$\langle \phi_{m\ell} | \delta_\sigma | \phi_{k\ell} \rangle = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 \int_0^\infty \phi_{m\ell}(r) e^{-\sigma^2 r^2} \phi_{k\ell}(r) dr. \quad (2.15)$$

For $\ell > 0$ the factor $r^{2\ell+2}$ in the integrand suppresses the contribution near the origin; in the limit $\sigma \rightarrow \infty$ the integral vanishes for $\ell > 0$ and reproduces $|\phi_{m\ell}(0)\phi_{k\ell}(0)|$ for $\ell = 0$, recovering the unregularised Dirac-delta result.

Remark. The signs $(-1)^{m+k}$ in Eqs. (2.13) and (2.14) encode the alternating parity of the Laguerre polynomials at the origin and are essential for obtaining the correct off-diagonal Cornell-potential matrix elements. In practice all of these closed-form expressions are evaluated to high precision via the `mpmath` library and verified against direct numerical quadrature.

2.1.5 The three channel Hamiltonian matrices

Assembling Eq. (2.11) channel by channel with the parameters of Table 2.1 and $\omega = 1.2 \text{ fm}^{-1}$ produces the following three 4×4 matrices (in units of fm^{-1} , with conversion to

MeV via $\hbar c = 197.326 \text{ MeV} \cdot \text{fm}$):

$$H^{(1S_0)} = \begin{pmatrix} 0.9431 & -0.8733 & -0.7690 & -0.5601 \\ -0.8733 & 3.3365 & -0.5646 & -0.8648 \\ -0.7690 & -0.5646 & 5.4382 & -0.1566 \\ -0.5601 & -0.8648 & -0.1566 & 7.3451 \end{pmatrix}, \quad (2.16)$$

$$H^{(3S_1)} = \begin{pmatrix} 1.0946 & -0.7114 & -0.6111 & -0.4112 \\ -0.7114 & 3.5406 & -0.3910 & -0.6989 \\ -0.6111 & -0.3910 & 5.6122 & -0.0119 \\ -0.4112 & -0.6989 & -0.0119 & 7.5104 \end{pmatrix}, \quad (2.17)$$

$$H^{(1P_1)} = \begin{pmatrix} 2.8561 & -0.2395 & -0.3827 & -0.2282 \\ -0.2395 & 4.9190 & -0.0373 & -0.5097 \\ -0.3827 & -0.0373 & 6.8114 & 0.4058 \\ -0.2282 & -0.5097 & 0.4058 & 8.6034 \end{pmatrix}. \quad (2.18)$$

Each matrix is real and symmetric, a consequence of the Hermiticity of the underlying operator and the choice of real-valued HO basis functions. The 1P_1 matrix lies higher in absolute energy because of the $\ell = 1$ centrifugal contribution to the diagonal; the spin-dependent shift between 1S_0 and 3S_1 is visible already in the difference between their diagonal entries.

Remark (Errata in the literature). We note that in the published matrices for 1P_1 [12] the (3, 3) element was reported as 7.5104, coinciding numerically with the (3, 3) entry of $H^{(3S_1)}$. Cross-checking against the variational eigenvalues and the diagonal pattern of the kinetic + centrifugal + confinement contributions identifies this as a copy-paste error; the correct value 8.6034 is used in Eq. (2.18).

Diagonalising each $H^{(\ell,S)}$ by classical means immediately yields the exact (within the truncated basis) energy spectrum of its channel. Doing so on a quantum computer, however, requires expressing the operator as a sum of Pauli strings acting on a register of qubits.

2.1.6 Pauli decomposition of a Hermitian matrix

A common building block for all three algorithms is the expansion of the channel Hamiltonian in the basis of Pauli strings. Any Hermitian operator H on n qubits can be written as a real linear combination of the 4^n Pauli strings $\{P_k\} \subset \{I, X, Y, Z\}^{\otimes n}$,

$$H = \sum_{k=1}^{4^n} c_k P_k, \quad c_k = \frac{1}{2^n} \text{Tr}(H P_k) \in \mathbb{R}. \quad (2.19)$$

The relation Eq. (2.19) follows from the orthogonality of Pauli strings under the Hilbert–Schmidt inner product, $\text{Tr}(P_j P_k) = 2^n \delta_{jk}$. In practice only a small subset of the 4^n coefficients are nonzero; for real symmetric Hamiltonians, all strings containing an odd number of Y factors vanish identically, leaving 3^n candidate terms.

Once H has been put in the form Eq. (2.19), each Pauli string can be implemented as a basis rotation on the quantum register followed by a diagonal measurement, and the expectation value $\langle H \rangle$ recovered as the weighted sum $\sum_k c_k \langle P_k \rangle$. The remaining question is which qubit register to use—i.e., how to associate physical basis states with bit-strings. The three source papers make two different choices, and we present each choice within the section that uses it.

2.2 Gallimore & Liao — Variational Quantum Eigensolver

Gallimore and Liao [11] apply the Variational Quantum Eigensolver (VQE) [5, 6] to a spinless Cornell-potential charmonium Hamiltonian truncated to $N = 3$ harmonic-oscillator basis states. Their construction maps the second-quantised one-body operator to qubits via the Jordan–Wigner transformation—exposing the problem on a register of three qubits with a one-particle constraint—and uses a two-parameter Unitary Coupled Cluster ansatz tailored to that constraint. In this section we reproduce their development, pointing out the parts that generalise without modification to the spin-dependent $N = 4$ matrices of Section 2.1. We close (Section 2.2.7) with the explicit extension required to apply VQE on equal footing with the other two methods later in the thesis.

2.2.1 The variational principle

For any Hermitian Hamiltonian H on a finite Hilbert space, every normalised state $|\psi\rangle$ obeys

$$E_0 \leq \langle \psi | H | \psi \rangle, \quad (2.20)$$

where E_0 is the smallest eigenvalue of H , with equality if and only if $|\psi\rangle$ lies in the corresponding eigenspace. Restricting the search to a family of *trial states* $\{|\psi(\boldsymbol{\theta})\rangle\}$ parameterised by a classical vector $\boldsymbol{\theta} \in \mathbb{R}^p$ gives the upper bound

$$E_0 \leq \min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}), \quad E(\boldsymbol{\theta}) \equiv \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle. \quad (2.21)$$

The VQE algorithm computes $E(\boldsymbol{\theta})$ on the quantum processor and minimises it on a classical computer; the resulting parameters $\boldsymbol{\theta}^*$ characterise an approximation $|\psi(\boldsymbol{\theta}^*)\rangle$ to the ground state. The accuracy of this approximation is controlled by the *expressibility* of the ansatz: if the parameterised manifold $\{|\psi(\boldsymbol{\theta})\rangle\}$ contains the true ground state then

the minimum of Eq. (2.21) coincides with E_0 ; otherwise it provides a (variational) upper bound.

The remaining ingredients are (i) a qubit encoding of H , (ii) a choice of ansatz, (iii) a procedure for measuring $E(\boldsymbol{\theta})$ from circuit shots, and (iv) a classical optimiser for $\boldsymbol{\theta}$. The next four subsections cover them in turn.

2.2.2 Jordan–Wigner encoding

The encoding chosen by Gallimore and Liao originates in quantum chemistry. We treat the HO basis functions as orbitals occupied by a single charm–anticharm pair and introduce a fermionic creation operator $a_n^\dagger$ for each orbital, with the canonical anticommutation relations

$$\{a_m, a_n^\dagger\} = \delta_{mn}, \quad \{a_m, a_n\} = \{a_m^\dagger, a_n^\dagger\} = 0. \quad (2.22)$$

The channel Hamiltonian is then the second-quantised one-body operator

$$\widehat{H}^{(\ell, S)} = \sum_{m, n=0}^{N-1} H_{mn}^{(\ell, S)} a_m^\dagger a_n, \quad (2.23)$$

acting on the Fock space of N orbitals; physical states are those containing exactly one excitation, i.e. the N states $\{a_n^\dagger |\text{vac}\rangle\}_{n=0}^{N-1}$. Restricting to this single-particle sector reproduces the original $N \times N$ matrix problem.

The Jordan–Wigner transformation. The Jordan–Wigner (JW) transformation [2] maps each orbital to a dedicated qubit and faithfully reproduces the fermionic algebra Eq. (2.22) in terms of Pauli operators,

$$a_n^\dagger = \frac{1}{2} \left(\prod_{j < n} Z_j \right) (X_n - i Y_n), \quad (2.24)$$

$$a_n = \frac{1}{2} \left(\prod_{j < n} Z_j \right) (X_n + i Y_n). \quad (2.25)$$

The trailing string of Z operators on lower-indexed qubits encodes the fermionic sign that arises when a creation operator anticommutes past already-occupied orbitals. With this mapping, N orbitals require N qubits, of which only $\binom{N}{1} = N$ computational-basis states are physical (for $N = 4$, these are $|0001\rangle, |0010\rangle, |0100\rangle, |1000\rangle$).

Pauli forms of the one-body operators. Substituting Eqs. (2.24) and (2.25) into the bilinears $a_m^\dagger a_n$ yields concrete Pauli strings. The number operator at orbital n is diagonal,

$$a_n^\dagger a_n = \frac{1}{2} (I - Z_n). \quad (2.26)$$

Adjacent hopping ($|m - n| = 1$, e.g. between orbitals n and $n + 1$) gives the familiar $XX+YY$ pattern,

$$a_n^\dagger a_{n+1} + a_{n+1}^\dagger a_n = \frac{1}{2} (X_n X_{n+1} + Y_n Y_{n+1}), \quad (2.27)$$

while non-adjacent hops pick up the JW Z -string,

$$a_n^\dagger a_{n+2} + a_{n+2}^\dagger a_n = \frac{1}{2} (X_n Z_{n+1} X_{n+2} + Y_n Z_{n+1} Y_{n+2}), \quad (2.28)$$

$$a_n^\dagger a_{n+3} + a_{n+3}^\dagger a_n = \frac{1}{2} (X_n Z_{n+1} Z_{n+2} X_{n+3} + Y_n Z_{n+1} Z_{n+2} Y_{n+3}). \quad (2.29)$$

Combining these expressions with the coefficients $H_{mn}^{(\ell,S)}$ from Section 2.1.5 produces a Pauli-string decomposition of $\widehat{H}^{(\ell,S)}$.

The $N = 4$ decomposition in full. For $N = 3$ orbitals (Gallimore and Liao's original setting) the diagonals contribute one identity and three single- Z terms, the three off-diagonals contribute pairs of $XX+YY$ or $XZX+YZY$ strings, giving ten Pauli terms in total. For the $N = 4$ extension used in this thesis, the same construction is carried out for the four orbitals on four qubits. Denoting the channel matrix elements $H_{mn} \equiv H_{mn}^{(\ell,S)}$ for brevity, and using the real symmetry $H_{mn} = H_{nm}$ to combine each off-diagonal pair $H_{mn}(a_m^\dagger a_n + a_n^\dagger a_m)$, the explicit Pauli expansion is

$$\begin{aligned} \widehat{H}^{(\ell,S)} = & \underbrace{\frac{1}{2} (H_{00} + H_{11} + H_{22} + H_{33}) I}_{\text{identity}} \\ & \underbrace{-\frac{1}{2} H_{00} Z_0 - \frac{1}{2} H_{11} Z_1 - \frac{1}{2} H_{22} Z_2 - \frac{1}{2} H_{33} Z_3}_{\text{4 diagonal } Z \text{ terms}} \\ & \underbrace{+\frac{1}{2} H_{01} (X_0 X_1 + Y_0 Y_1) + \frac{1}{2} H_{12} (X_1 X_2 + Y_1 Y_2) + \frac{1}{2} H_{23} (X_2 X_3 + Y_2 Y_3)}_{\text{3 adjacent pairs}} \\ & \underbrace{+\frac{1}{2} H_{02} (X_0 Z_1 X_2 + Y_0 Z_1 Y_2) + \frac{1}{2} H_{13} (X_1 Z_2 X_3 + Y_1 Z_2 Y_3)}_{\text{2 skip-1 pairs}} \\ & \underbrace{+\frac{1}{2} H_{03} (X_0 Z_1 Z_2 X_3 + Y_0 Z_1 Z_2 Y_3)}_{\text{1 skip-2 pair}}. \end{aligned} \quad (2.30)$$

Counting strings gives one identity, four single- Z terms, three pairs of adjacent $XX+YY$ strings (6 strings), two pairs of skip-1 $XZX+YZY$ strings (4 strings), and one pair of skip-2 $XZZX+YZZY$ strings (2 strings), totalling $1 + 4 + 6 + 4 + 2 = 17$ Pauli operators acting on four qubits.

Remark (Structure of the coefficients). Each Pauli string in Eq. (2.30) carries a coefficient that is either a single diagonal entry H_{nn} (for the Z_n terms) or a single off-diagonal entry H_{mn} (for each member of a pair). Crucially, the *operator structure* of the Pauli decomposition—which strings appear, and with what relative signs—is inde-

pendent of which channel we work with: the channel-specific information enters only through the numerical coefficients. The same 17 Pauli strings of Eq. (2.30) therefore describe $\widehat{H}^{(1S_0)}$, $\widehat{H}^{(3S_1)}$, and $\widehat{H}^{(1P_1)}$, with the substitutions $H_{mn} \rightarrow H_{mn}^{(1S_0)}$, $H_{mn}^{(3S_1)}$, and $H_{mn}^{(1P_1)}$ from Eqs. (2.16) to (2.18).

2.2.3 Unitary Coupled Cluster ansatz

Within the single-particle sector, the trial state must remain particle-number-conserving. Gallimore and Liao construct an ansatz from that constraint by exponentiating particle-conserving generators. Starting from the reference state $|0 \cdots 01\rangle$ (one excitation in the lowest orbital), the Unitary Coupled Cluster (UCC) operator is

$$U_{\text{UCC}}(\theta_1, \dots, \theta_{N-1}) = \exp \left\{ \sum_{k=1}^{N-1} \theta_k (a_k^\dagger a_0 - a_0^\dagger a_k) \right\}. \quad (2.31)$$

Each generator $a_k^\dagger a_0 - a_0^\dagger a_k$ is anti-Hermitian and preserves particle number, so U_{UCC} is unitary and maps single-particle states to single-particle states. With $N - 1$ independent parameters the operator generates the most general real superposition in the single-particle sector,

$$|\psi(\boldsymbol{\theta})\rangle = \cos \theta_{N-1} |0 \cdots 01\rangle + \sin \theta_{N-1} \cos \theta_{N-2} |0 \cdots 010\rangle + \cdots + \sin \theta_{N-1} \cdots \sin \theta_1 |10 \cdots 0\rangle, \quad (2.32)$$

i.e. the $\mathbb{S}^{N-1}$ spherical parameterisation lifted into the single-particle sector. For $N = 3$ this is the two-parameter ansatz of Gallimore and Liao [11]; for $N = 4$ it provides three parameters and exhausts the real single-particle subspace.

Circuit implementation via Givens rotations. A natural circuit-level realisation of Eq. (2.32) uses a chain of *Givens rotations* between adjacent orbitals. Within the single-particle subspace of two adjacent qubits $(j, j + 1)$, a Givens rotation is the orthogonal 2×2 rotation acting on the basis $\{|01\rangle, |10\rangle\}$:

$$G_{j,j+1}(\theta) : \begin{cases} |00\rangle \mapsto |00\rangle, \\ |01\rangle \mapsto \cos \theta |01\rangle + \sin \theta |10\rangle, \\ |10\rangle \mapsto -\sin \theta |01\rangle + \cos \theta |10\rangle, \\ |11\rangle \mapsto |11\rangle, \end{cases} \quad (2.33)$$

which is precisely the exponential $\exp\{\theta (a_{j+1}^\dagger a_j - a_j^\dagger a_{j+1})\}$ restricted to the two qubits in question. A standard hardware decomposition in terms of one- and two-qubit native gates is

$$G_{j,j+1}(\theta) = \text{CNOT}_{j+1 \rightarrow j} \left(R_y(2\theta) \right)_{j+1}^{\text{ctrl } j} \text{CNOT}_{j+1 \rightarrow j}, \quad (2.34)$$

i.e. two CNOTs and one controlled- R_y , giving the Givens block constant depth.

Acting on the reference state $|0 \cdots 01\rangle$ (qubit 0 occupied, all others empty) with a chain of Givens rotations between adjacent qubits produces the spherical-coordinate expansion of Eq. (2.32) directly:

$$|0 \cdots 01\rangle \xrightarrow{G_{0,1}(\theta_{N-1})} \cdots \xrightarrow{G_{N-2,N-1}(\theta_1)} |\psi(\boldsymbol{\theta})\rangle,$$

with each rotation transferring the particle one orbital further from the reference. The resulting circuit for $N = 4$ is shown in Fig. 2.1; its depth is $N - 1$ Givens blocks, growing linearly in the basis size.

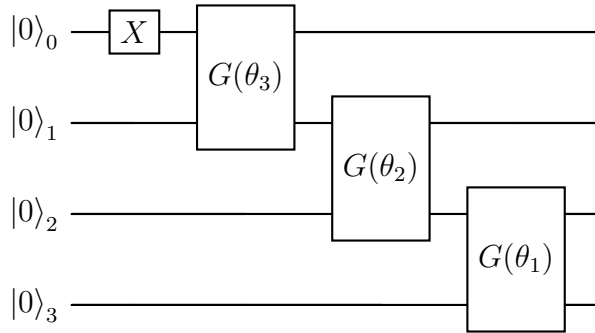


Figure 2.1: Three-parameter UCC ansatz for the $N = 4$ Jordan–Wigner encoding. An X gate prepares the reference state $|0001\rangle$ from $|0000\rangle$; three Givens rotations $G_{j,j+1}(\theta_{N-1-j})$, each decomposed into two CNOTs and a controlled- R_y via Eq. (2.34), propagate the single excitation through the chain and generate the spherical superposition of Eq. (2.32). Particle number is exactly conserved by construction.

The circuit has 1 single-qubit X gate, 3 controlled- R_y gates, and 6 CNOTs in total—comparable in depth to the Woloshyn ansatz of Fig. 2.3 despite using twice as many qubits, because the single-particle constraint sharply restricts the relevant subspace.

2.2.4 Energy measurement on a quantum processor

Given the ansatz $|\psi(\boldsymbol{\theta})\rangle$ and the Pauli decomposition $H = \sum_k c_k P_k$ from Eq. (2.19), the energy estimator is linear in the Pauli expectation values,

$$\hat{E}(\boldsymbol{\theta}) = \sum_k c_k \langle P_k \rangle_{\boldsymbol{\theta}}, \quad (2.35)$$

where each $\langle P_k \rangle_{\boldsymbol{\theta}}$ is estimated separately by preparing $|\psi(\boldsymbol{\theta})\rangle$ many times and measuring P_k .

A measurement in the computational basis directly gives access to $\langle Z_i \rangle$, but X and Y observables require a basis rotation applied before the measurement. The rotations are obtained from the eigenbasis identities $X = HZH$ and $Y = (SH)Z(HS^\dagger)$, giving the

standard recipe

$$\langle X_i \rangle = \text{measure } Z_i \text{ after applying } H \text{ on qubit } i, \quad (2.36)$$

$$\langle Y_i \rangle = \text{measure } Z_i \text{ after applying } S^\dagger H \text{ on qubit } i. \quad (2.37)$$

Each Pauli string P_k is mapped to a diagonal observable by applying these single-qubit basis rotations to every non-identity factor before the final measurement.

Remark (Shot noise). With a finite number N_{shots} of repetitions per circuit, the estimator Eq. (2.35) fluctuates around its true value with a statistical uncertainty $\sigma_{\hat{E}} = \sqrt{\sum_k c_k^2 \sigma_{P_k}^2 / N_{\text{shots}}}$, where $\sigma_{P_k}^2 \leq 1$ for any Pauli. The relative error therefore decreases only as $1/\sqrt{N_{\text{shots}}}$, which makes shot count an important resource and motivates the choice of optimiser discussed next.

2.2.5 Classical optimisation

The cost function $\hat{E}(\boldsymbol{\theta})$ is, in general, non-convex and contaminated by shot noise. Two broad families of classical optimisers are available:

- **Gradient-free methods** (e.g. COBYLA, Nelder–Mead) only require evaluations of $\hat{E}$ and are robust to moderate statistical noise. They typically converge in tens to a few hundreds of iterations for low-dimensional parameter spaces.
- **Gradient-based methods**, which require partial derivatives of $\hat{E}$ at the cost of additional circuit evaluations. The *parameter-shift rule* (Section 2.3.6 below) provides exact analytic gradients for rotation gates, but each gradient component doubles the number of circuits.

Gallimore and Liao use COBYLA, which balances simplicity, noise tolerance, and convergence speed for low-parameter ansätze. Figure 2.2 recalls the resulting hybrid loop already introduced in Chapter 1: the classical optimiser proposes a parameter update, the quantum processor evaluates $\hat{E}(\boldsymbol{\theta})$, and the cycle repeats until convergence.

2.2.6 Excited states via orthogonalisation

VQE in the form above targets only the ground state. To reach excited levels, Gallimore and Liao constrain the trial state to be orthogonal to the previously obtained eigenstates. Suppose the ground state $|\psi_0(\boldsymbol{\theta}_0^*)\rangle$ has been determined; the first excited state is then obtained by minimising

$$E_1 = \min_{\boldsymbol{\theta}: \langle \psi_0 | \psi(\boldsymbol{\theta}) \rangle = 0} \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle. \quad (2.38)$$

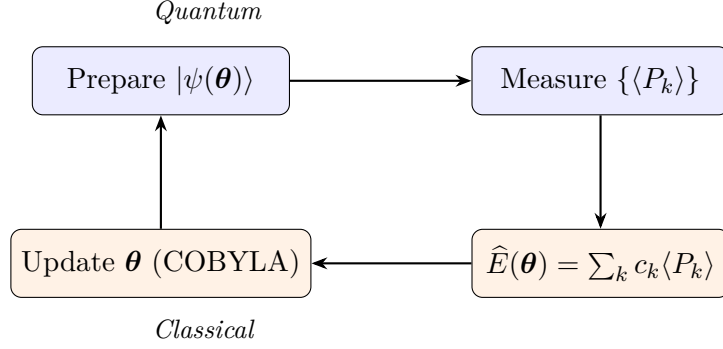


Figure 2.2: The VQE hybrid loop specialised to charmonium spectroscopy. The quantum processor prepares the UCC ansatz of Eq. (2.32), measures the Pauli expectation values in Eq. (2.35), and returns the energy estimate to a classical optimiser.

For the two-parameter UCC ansatz at $N = 3$ this constrained optimisation is one-dimensional: the orthogonal complement of $|\psi_0\rangle$ inside the ansatz manifold is parameterised by a single angle γ , and the optimisation reduces to a one-dimensional scan [11]. The orthogonal trial state takes the form

$$|\psi_1(\gamma)\rangle = \cos \gamma |\psi_0^{\perp,a}\rangle + \sin \gamma |\psi_0^{\perp,b}\rangle, \quad (2.39)$$

where $|\psi_0^{\perp,a}\rangle, |\psi_0^{\perp,b}\rangle$ are two fixed states orthogonal to $|\psi_0\rangle$ that span the rest of the real ansatz manifold. Iterating the procedure, each new excited state is constrained to be orthogonal to all previously obtained states.

Remark (Error propagation). A useful property of the orthogonalisation method is that errors in the ground-state estimate propagate in a bounded way: if the variational ground state has an error $\delta_0 = E_0^{\text{VQE}} - E_0$, the first excited state estimate E_1^{VQE} satisfies $|E_1^{\text{VQE}} - E_1| \leq \delta_0$ [11], so the accuracy does not deteriorate as we move up the spectrum.

2.2.7 Extension to the spin-dependent N=4 case

The Gallimore–Liao construction was originally applied to a *spin-independent* Cornell potential truncated to $N = 3$, producing a 3×3 Hamiltonian, a JW Hamiltonian on three qubits, and a two-parameter UCC ansatz. In this thesis we want to compare VQE with VQITE (Section 2.3) and TTITE (Section 2.4) on the same physical system—the spin-dependent 1S_0 , 3S_1 , 1P_1 matrices of Section 2.1.5. The Gallimore–Liao construction extends in two respects to make this possible.

- **Spin dependence.** The Pauli-string structure of Eq. (2.23) is unchanged when the Hamiltonian matrix carries an extra term from the contact interaction Eq. (2.3): the matrix elements $H_{mn}^{(\ell,S)}$ themselves shift between channels, but the second-quantised expression and the resulting JW Pauli decomposition are formally identical. The spin enters only through the choice of coefficients.

- **Basis size.** Moving from $N = 3$ to $N = 4$ adds one more orbital, hence one more qubit and one more UCC parameter. Equation (2.29) contributes the new long-range hops, yielding the full 17-term Pauli expansion of Eq. (2.30). The UCC operator gains a third generator $a_3^\dagger a_0 - a_0^\dagger a_3$ and a third parameter, producing the three-parameter form of Eq. (2.32).

The resulting four-qubit VQE algorithm is a faithful extension of Gallimore and Liao’s procedure; we are not introducing new ingredients, only enlarging the basis and admitting the spin-dependent coefficients already present in the underlying matrix. With this extension in place, VQE can be applied to exactly the same 4×4 matrices used by VQITE and TTITE in the next two sections, enabling the apples-to-apples comparison reported in the next chapter.

2.3 Woloshyn — Variational Imaginary-Time Evolution

Woloshyn [12] simulates the same spin-dependent charmonium system studied in this thesis—the 1S_0 , 3S_1 , and 1P_1 channel matrices of Section 2.1.5—using the Variational Imaginary-Time Evolution (VQITE) algorithm of McArdle et al. [16] and Yuan et al. [17]. Unlike VQE, VQITE locates the ground state by tracking the (non-unitary) imaginary-time Schrödinger equation through a parameterised ansatz manifold, deriving classical equations of motion from a variational principle. Woloshyn uses the compact direct mapping for the qubit encoding—two qubits for four basis states—together with a three-parameter R_y -based ansatz, and also computes the M1 and E1 radiative transition amplitudes between channels. We reproduce all of this material below.

2.3.1 Imaginary-time evolution as a ground-state projector

The imaginary-time Schrödinger equation is obtained from its real-time counterpart by the Wick rotation $t \mapsto -i\tau$, with $\tau \geq 0$, giving

$$\frac{d|\psi(\tau)\rangle}{d\tau} = -H|\psi(\tau)\rangle. \quad (2.40)$$

Its (unnormalised) solution is $e^{-H\tau}|\psi(0)\rangle$. Expanding the initial state in the energy eigenbasis, $|\psi(0)\rangle = \sum_k a_k |E_k\rangle$, gives

$$e^{-H\tau}|\psi(0)\rangle = \sum_k a_k e^{-E_k\tau} |E_k\rangle. \quad (2.41)$$

Factoring out the ground-state contribution and renormalising,

$$|\psi(\tau)\rangle = \frac{a_0 |E_0\rangle + \sum_{k>0} a_k e^{-(E_k-E_0)\tau} |E_k\rangle}{\sqrt{|a_0|^2 + \sum_{k>0} |a_k|^2 e^{-2(E_k-E_0)\tau}}} \xrightarrow{\tau \rightarrow \infty} \frac{a_0}{|a_0|} |E_0\rangle, \quad (2.42)$$

provided $a_0 = \langle E_0 | \psi(0) \rangle \neq 0$. The excited-state amplitudes are exponentially suppressed at a rate set by the gap $E_k - E_0$, so the algorithm reaches the ground state in a number of imaginary-time steps proportional to $1/(E_1 - E_0)$.

To preserve the norm dynamically, it is convenient to subtract the instantaneous energy, $E_\tau \equiv \langle \psi(\tau) | H | \psi(\tau) \rangle$, from H . The *normalised* imaginary-time equation

$$\frac{d|\psi(\tau)\rangle}{d\tau} = -(H - E_\tau) |\psi(\tau)\rangle \quad (2.43)$$

preserves the norm of $|\psi\rangle$ and has the same long- τ limit as Eq. (2.40), but is more convenient to discretise variationally.

Remark (Non-unitarity). The operator $e^{-H\tau}$ is Hermitian and positive definite, but it is not unitary: $(e^{-H\tau})^\dagger e^{-H\tau} = e^{-2H\tau} \neq I$. Quantum circuits, by contrast, are unitary by construction. Implementing Eq. (2.43) on a quantum processor therefore requires an indirect strategy; VQITE projects the exact non-unitary dynamics onto a unitary parameterised submanifold, while Section 2.4 will encode it probabilistically using ancillas.

2.3.2 Direct (compact) encoding

Woloshyn adopts the most economical qubit encoding compatible with the $N = 4$ basis: each HO basis function is assigned to one computational-basis state of a two-qubit register via the binary expansion of its index,

$$|n\rangle_{\text{HO}} \mapsto |\text{bin}(n)\rangle_{\text{qubit}}, \quad n = 0, 1, 2, 3. \quad (2.44)$$

Explicitly,

$$|0\rangle \mapsto |00\rangle, \quad |1\rangle \mapsto |01\rangle, \quad |2\rangle \mapsto |10\rangle, \quad |3\rangle \mapsto |11\rangle. \quad (2.45)$$

For $N = 4$ this requires only $\lceil \log_2 N \rceil = 2$ qubits, and every computational-basis state of the register corresponds to a physical HO state—there are no spurious states and no symmetry constraint on the ansatz.

The channel Hamiltonian $H^{(\ell, S)}$ is then literally a 4×4 matrix acting on this register, and its Pauli decomposition follows directly from Eq. (2.19). Applying that formula to the 1S_0 matrix of Eq. (2.16) yields, for example,

$$H^{(^1S_0)} = c_{II} II + c_{IZ} IZ + c_{ZI} ZI + c_{ZZ} ZZ + c_{IX} IX + c_{XI} XI + c_{ZX} ZX + c_{XZ} XZ + c_{XX} XX + c_{YY} YY, \quad (2.46)$$

i.e. ten Pauli terms, consistent with the absence of strings containing an odd number of Y factors for a real-symmetric operator. The numerical coefficients $\{c_k\}$ are obtained by the trace formula of Eq. (2.19) and are real.

Example (Identity coefficient). Setting $P_k = I \otimes I$ in Eq. (2.19) gives the energy offset

$$c_{II} = \frac{1}{4} \text{Tr } H^{(1S_0)} = \frac{1}{4}(0.9431 + 3.3365 + 5.4382 + 7.3451) \approx 4.266,$$

which simply records the trace of the channel matrix and contributes only a constant shift to the eigenvalues.

The direct encoding has two clear advantages over the JW encoding of Section 2.2.2: it uses the minimal number of qubits, and every computational-basis state is physical, so the variational ansatz need not respect any external constraint. Its main drawback is that the resulting Pauli strings carry no obvious physical interpretation—they are simply the coefficients of an abstract 4×4 matrix.

2.3.3 Three-parameter variational ansatz

For the two-qubit direct encoding, a state of the form

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle, \quad c_n \in \mathbb{R}, \quad \sum_n c_n^2 = 1, \quad (2.47)$$

has three independent real degrees of freedom (four amplitudes minus the normalisation). Because the channel Hamiltonians of Eqs. (2.16) to (2.18) are real and symmetric, their eigenstates can be chosen real, and a three-parameter circuit producing the most general real state of the form Eq. (2.47) is therefore expressible enough. Woloshyn’s circuit applies two single-qubit R_y rotations, a CNOT, and a third R_y :

$$U(\boldsymbol{\theta}) = [I \otimes R_y(\theta_2)] \text{CNOT}_{0 \rightarrow 1} [R_y(\theta_1) \otimes R_y(\theta_0)], \quad (2.48)$$

with $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)$. The corresponding circuit is shown in Fig. 2.3.

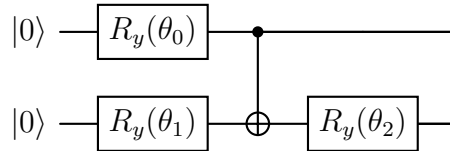


Figure 2.3: Three-parameter ansatz used with the direct encoding. The two initial R_y rotations create an arbitrary product state, the CNOT entangles the qubits, and the final R_y provides the third independent degree of freedom required to reach a general real four-amplitude state.

Acting on $|00\rangle$, this circuit yields amplitudes

$$|\psi(\boldsymbol{\theta})\rangle = \begin{pmatrix} c_0(c_1c_2 - s_1s_2) \\ s_0(s_1c_2 - c_1s_2) \\ c_0(c_1s_2 + s_1c_2) \\ s_0(s_1s_2 + c_1c_2) \end{pmatrix}, \quad c_n = \cos(\theta_n/2), \quad s_n = \sin(\theta_n/2), \quad (2.49)$$

which can be verified to cover the unit sphere in $\mathbb{R}^4$ as $\boldsymbol{\theta}$ ranges over a domain of size $4\pi^3$, modulo redundancies.

2.3.4 The McLachlan variational principle

VQITE replaces the exact evolution Eq. (2.43) by a *parameterised* trajectory $|\psi(\boldsymbol{\theta}(\tau))\rangle$ on the ansatz manifold of Section 2.3.3. The parameters $\boldsymbol{\theta}(\tau)$ are evolved so that the parameterised state tracks the exact one as closely as possible. We adopt the principle of McLachlan [18], which minimises at each instant the distance between the actual time-derivative of the trial state and the right-hand side of Eq. (2.43):

$$\delta_{\dot{\boldsymbol{\theta}}} \left\| \frac{\partial}{\partial \tau} |\psi(\boldsymbol{\theta}(\tau))\rangle + (H - E_\tau) |\psi(\boldsymbol{\theta}(\tau))\rangle \right\|^2 = 0. \quad (2.50)$$

Here the variation is taken with respect to the time derivative $\dot{\boldsymbol{\theta}} \equiv d\boldsymbol{\theta}/d\tau$, treated as an independent real vector. The variation reduces, in five short steps, to the linear system

$$\sum_j A_{ij} \dot{\theta}_j = C_i, \quad i = 1, \dots, p, \quad (2.51)$$

with matrix and vector elements

$$A_{ij} = \Re \left[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle \right], \quad (2.52)$$

$$C_i = -\Re \left[\langle \partial_i \psi | H | \psi \rangle \right] + \Re \left[\langle \partial_i \psi | \psi \rangle \right] E_\tau. \quad (2.53)$$

We work through the derivation in detail since the same building blocks will reappear when we discuss the metric tensor and the parameter-shift rule below.

Step 1 — expand the time derivative. The ansatz state depends on τ only through the parameters, so by the chain rule

$$\partial_\tau |\psi(\boldsymbol{\theta}(\tau))\rangle = \sum_i \dot{\theta}_i |\partial_i \psi\rangle, \quad |\partial_i \psi\rangle \equiv \frac{\partial |\psi(\boldsymbol{\theta})\rangle}{\partial \theta_i}. \quad (2.54)$$

Step 2 — write the residual vector. Denote $K \equiv H - E_\tau$ for brevity. The vector inside the norm in Eq. (2.50) is

$$|R\rangle = \sum_i \dot{\theta}_i |\partial_i \psi\rangle + K |\psi\rangle. \quad (2.55)$$

Step 3 — compute its squared norm. Expanding $\langle R|R\rangle$,

$$\|R\|^2 = \sum_{i,j} \dot{\theta}_i \dot{\theta}_j \langle \partial_i \psi | \partial_j \psi \rangle + \sum_i \dot{\theta}_i \langle \partial_i \psi | K | \psi \rangle + \sum_j \dot{\theta}_j \langle \psi | K | \partial_j \psi \rangle + \langle \psi | K^2 | \psi \rangle.$$

The two middle sums are complex conjugates of each other (since K is Hermitian and $\dot{\theta}_i \in \mathbb{R}$); they combine into a single real part. In the first sum the antisymmetric imaginary part of $\langle \partial_i \psi | \partial_j \psi \rangle$ vanishes against the symmetric $\dot{\theta}_i \dot{\theta}_j$, leaving only the real part. Hence

$$\|R\|^2 = \sum_{i,j} \dot{\theta}_i \dot{\theta}_j \Re \langle \partial_i \psi | \partial_j \psi \rangle + 2 \sum_i \dot{\theta}_i \Re \langle \partial_i \psi | K | \psi \rangle + \|K | \psi \rangle\|^2. \quad (2.56)$$

Step 4 — vary with respect to $\dot{\theta}_k$. The final term in Eq. (2.56) is independent of $\dot{\theta}$. Differentiating the remainder with respect to $\dot{\theta}_k$ and setting the derivative to zero,

$$\frac{\partial \|R\|^2}{\partial \dot{\theta}_k} = 2 \sum_j \Re \langle \partial_k \psi | \partial_j \psi \rangle \dot{\theta}_j + 2 \Re \langle \partial_k \psi | K | \psi \rangle = 0, \quad (2.57)$$

which is already linear in $\dot{\theta}$, matching the structure of Eq. (2.51) with the *bare* entries

$$A_{kj}^{(0)} = \Re \langle \partial_k \psi | \partial_j \psi \rangle, \quad (2.58)$$

$$C_k = -\Re \langle \partial_k \psi | K | \psi \rangle = -\Re \langle \partial_k \psi | H | \psi \rangle + E_\tau \Re \langle \partial_k \psi | \psi \rangle. \quad (2.59)$$

The right-hand side of Eq. (2.59) is precisely Eq. (2.53), so the vector C_i is recovered at this step already.

Step 5 — project to the Fubini–Study form. The bare metric Eq. (2.58) misses one subtlety: the parameterised state has a redundant global-phase degree of freedom— $|\psi(\boldsymbol{\theta})\rangle$ and $e^{i\phi(\boldsymbol{\theta})} |\psi(\boldsymbol{\theta})\rangle$ represent the same physical state, but their tangent vectors differ by an arbitrary multiple of $|\psi\rangle$. To remove this gauge redundancy we project the tangent vectors orthogonal to $|\psi\rangle$,

$$|\partial_i \psi\rangle_\perp \equiv \left(I - |\psi\rangle \langle \psi| \right) |\partial_i \psi\rangle = |\partial_i \psi\rangle - |\psi\rangle \langle \psi | \partial_i \psi \rangle, \quad (2.60)$$

and use $|\partial_i \psi\rangle_\perp$ in place of $|\partial_i \psi\rangle$ throughout. A direct computation using $\langle \psi | \psi \rangle = 1$ gives

$$\langle \partial_i \psi |_\perp | \partial_j \psi |_\perp \rangle = \langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle,$$

whose real part is exactly Eq. (2.52). The same projection on $|\partial_k \psi\rangle$ in C_k leaves it unchanged, because $\langle \psi | K | \psi \rangle = E_\tau - E_\tau = 0$ already; only A acquires the correction. For ansatz circuits that preserve the global phase of $|\psi\rangle$ (which is the case for the construction of Section 2.3.3) the second term in C_i vanishes and C_i reduces to a real-valued energy gradient,

$$C_i = -\frac{1}{2} \frac{\partial E(\boldsymbol{\theta})}{\partial \theta_i}. \quad (2.61)$$

The matrix A is precisely the (real) Fubini–Study quantum metric on the ansatz manifold; it is real, symmetric, and positive semidefinite, and it plays the same role for variational quantum evolution as the Fisher information matrix in classical statistics [19].

2.3.5 The quantum metric tensor in practice

The geometric interpretation of A_{ij} is illuminating, but more useful for implementation is its expression as a curvature of the *fidelity surface*

$$f(\boldsymbol{\theta}, \boldsymbol{\theta}') \equiv |\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}') \rangle|^2. \quad (2.62)$$

Expanding around $\boldsymbol{\theta}' = \boldsymbol{\theta}$ to second order yields

$$f(\boldsymbol{\theta}, \boldsymbol{\theta} + \boldsymbol{\delta}) = 1 - \sum_{ij} A_{ij} \delta_i \delta_j + O(\delta^3), \quad (2.63)$$

so that the metric coincides with the leading curvature of the fidelity landscape. This identification is the basis of the standard finite-difference estimator: from Eq. (2.63),

$$A_{ij} \approx -\frac{1}{4\varepsilon^2} \left[f(\boldsymbol{\theta} + \varepsilon e_i + \varepsilon e_j) - f(\boldsymbol{\theta} + \varepsilon e_i - \varepsilon e_j) - f(\boldsymbol{\theta} - \varepsilon e_i + \varepsilon e_j) + f(\boldsymbol{\theta} - \varepsilon e_i - \varepsilon e_j) \right], \quad (2.64)$$

with e_i the unit vector in the i -direction. Each of the four fidelities is the absolute square of an overlap between two ansatz states, which can be estimated on a quantum processor either by the swap test or by the inverse-ansatz protocol described later in Section 2.3.9. For the three-parameter ansatz used here, the symmetric metric requires $p(p+1)/2 = 6$ independent entries, each costing four overlap measurements at a small step $\varepsilon \sim 10^{-3}$ – 10^{-4} .

2.3.6 Energy gradient and the parameter-shift rule

The vector C_i in Eq. (2.61) is a derivative of an expectation value with respect to a circuit parameter. For ansatzes composed of single-qubit rotations of the form $R_\mu(\theta) = e^{-i\theta \mu/2}$ with $\mu \in \{X, Y, Z\}$, such derivatives admit a remarkable closed-form expression—the

parameter-shift rule [20, 21]:

$$\frac{\partial \langle H \rangle_{\boldsymbol{\theta}}}{\partial \theta_i} = \frac{1}{2} \left[\langle H \rangle_{\boldsymbol{\theta} + (\pi/2) \mathbf{e}_i} - \langle H \rangle_{\boldsymbol{\theta} - (\pi/2) \mathbf{e}_i} \right]. \quad (2.65)$$

The remarkable feature of Eq. (2.65) is that it is *exact*—unlike a finite-difference derivative, the formula returns the true partial derivative regardless of the step size. The proof, which we sketch here, illustrates the special role played by Pauli rotations.

Remark (Sketch of the parameter-shift derivation). Consider a circuit where the parameter θ_i appears only inside a rotation $R_{\mu}(\theta_i) = e^{-i\theta_i \mu/2}$. Since $\mu^2 = I$, we have $R_{\mu}(\theta_i) = \cos(\theta_i/2) I - i \sin(\theta_i/2) \mu$, so the dependence of R_{μ} —and hence of $|\psi(\boldsymbol{\theta})\rangle$ —on θ_i is restricted to the linear span of $\{1, \cos \theta_i, \sin \theta_i\}$. Therefore the expectation value $\langle H \rangle_{\boldsymbol{\theta}}$ is a real-valued trigonometric polynomial of period 2π in θ_i , of the form $a \cos \theta_i + b \sin \theta_i + c$. Setting the right-hand side of Eq. (2.65) equal to $b \cos \theta_i - a \sin \theta_i$, which is exactly the derivative of $a \cos \theta_i + b \sin \theta_i + c$, completes the proof.

The practical implication is that, for the R_y -based ansatz of Eq. (2.48), all partial derivatives required by VQITE—i.e. each component of C —can be evaluated on the quantum processor at a cost of two additional energy estimations per parameter. For the three-parameter ansatz used here, the cost is just six extra circuit evaluations per imaginary-time step.

2.3.7 Time stepping and regularisation

Combining Eqs. (2.51), (2.52) and (2.61) gives the equation of motion for $\boldsymbol{\theta}$:

$$\dot{\boldsymbol{\theta}} = A^{-1} C. \quad (2.66)$$

Discretising with the forward Euler scheme yields a simple update rule for the parameter vector after a small imaginary-time step $\Delta\tau$,

$$\boldsymbol{\theta}(\tau + \Delta\tau) = \boldsymbol{\theta}(\tau) + \Delta\tau A^{-1}(\boldsymbol{\theta}) C(\boldsymbol{\theta}). \quad (2.67)$$

Two practical issues arise. First, when two parameters become approximately degenerate (e.g., when the ansatz fails to be locally injective), the metric A becomes ill-conditioned. We add a small Tikhonov regulariser to restore numerical stability,

$$(A + \lambda I) \dot{\boldsymbol{\theta}} = C, \quad \lambda \sim 10^{-4}. \quad (2.68)$$

Second, the explicit Euler scheme is conditionally stable and accuracy is limited by the curvature of the energy landscape. Step sizes around $\Delta\tau \simeq 0.02$ in natural units of $1/\|H\|$ are sufficient to reach a stable ground state in 50–100 steps for the channel Hamiltonians

of Eqs. (2.16) to (2.18).

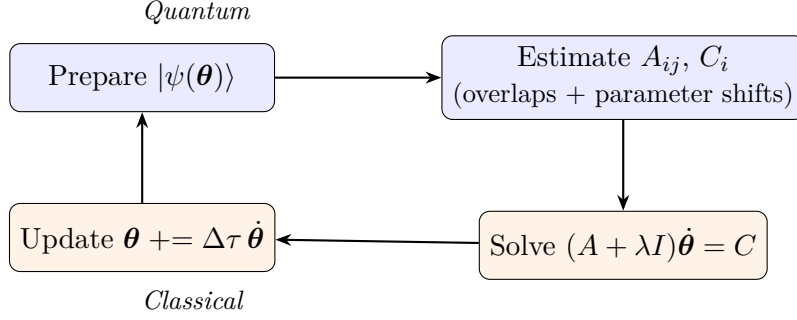


Figure 2.4: The VQITE hybrid loop. At each step the quantum processor estimates the metric tensor A via Eq. (2.64) and the energy gradient components of C via the parameter-shift rule Eq. (2.65); the classical computer solves the linear system Eq. (2.68) and advances the parameters in imaginary time.

2.3.8 Excited states by penalisation

VQITE projects onto the lowest-energy state with non-vanishing overlap with the initial state. To target higher levels we modify the effective Hamiltonian by adding penalty terms that suppress overlap with previously obtained states. Concretely, if $\{|\psi_k^*\rangle\}_{k=0}^{n-1}$ are already-computed eigenstates, we replace H by

$$H_{\text{eff}} = H + \alpha \sum_{k=0}^{n-1} |\psi_k^*\rangle \langle \psi_k^*|, \quad (2.69)$$

where α is a positive penalty large enough to shift the energies of the already-found states above the next eigenvalue. Running VQITE with H_{eff} then converges to the next-higher eigenstate. The energy gradient generalises to

$$\frac{\partial \langle H_{\text{eff}} \rangle}{\partial \theta_i} = \frac{\partial \langle H \rangle}{\partial \theta_i} + 2\alpha \sum_k \Re \left[\langle \psi_k^* | \psi \rangle \langle \partial_i \psi | \psi_k^* \rangle \right], \quad (2.70)$$

where the overlap correction is again accessible from short fidelity circuits. In practice α is chosen of the same order of magnitude as the typical gap between adjacent eigenvalues; we use $\alpha = 10$ in inverse-femtometre units throughout this thesis.

The combination of VQITE for the ground state with iterated penalisation provides access to a sequence of eigenpairs without ever requiring the diagonalisation of an exponentially large matrix on a classical computer.

2.3.9 Transition amplitudes

Woloshyn’s analysis goes beyond eigenvalues: it also computes radiative transition amplitudes between channels, which are the physical observables ultimately compared with

experiment. We reproduce the two relevant cases below.

Physical motivation. In the multipole expansion of the electromagnetic interaction of a quark–antiquark system, the leading single-photon transitions are classified by their selection rules [10]:

- *M1 transitions* flip the total spin without changing the orbital angular momentum ($\Delta S = 1, \Delta L = 0$). In the non-relativistic limit the amplitude reduces to a wavefunction overlap between the two spin partners, scaled by a numerical prefactor.
- *E1 transitions* change the orbital angular momentum by one unit without altering the spin ($\Delta S = 0, \Delta L = 1$). The amplitude is proportional to the radial matrix element of the position operator, $\langle \psi_f | r | \psi_i \rangle$.

The two paradigmatic examples for charmonium are $J/\psi \rightarrow \eta_c \gamma$ (M1) and $h_c \rightarrow \eta_c \gamma$ (E1). Computing their amplitudes therefore amounts to evaluating, respectively, the overlap $\langle \psi_f^{(^1S_0)} | \psi_i^{(^3S_1)} \rangle$ and the matrix element $\langle \psi_f^{(^1S_0)} | r | \psi_i^{(^1P_1)} \rangle$, where the states are the eigenvectors produced by VQITE (or by any of the other two algorithms).

M1 amplitudes: state overlaps. In the non-relativistic limit, the M1 transition amplitude between two states differing only in spin reduces to an overlap of their spatial wavefunctions,

$$|\mathcal{M}_{M1}|^2 \propto \left| \langle \psi_f^{\text{spatial}} | \psi_i^{\text{spatial}} \rangle \right|^2. \quad (2.71)$$

The two states are eigenvectors of different channel Hamiltonians (1S_0 and 3S_1) but live in the same HO basis, so their overlap is well-defined as a four-dimensional inner product of real coefficient vectors. Two complementary quantum-circuit techniques are used in Woloshyn’s analysis.

Inverse-ansatz protocol. If the two ansatz unitaries $U_i = U(\boldsymbol{\theta}_i^*)$ and $U_f = U(\boldsymbol{\theta}_f^*)$ have been determined—e.g., as the output of two VQITE runs—then the overlap can be read off from the all-zero outcome of the circuit $U_f^\dagger U_i |0\rangle^{\otimes n}$. Indeed,

$$\text{Prob}(\text{all zeros}) = \left| \langle 0 |^{\otimes n} U_f^\dagger U_i | 0 \rangle^{\otimes n} \right|^2 = \left| \langle \psi_f | \psi_i \rangle \right|^2. \quad (2.72)$$

This protocol requires no additional qubits, only an inverse-ansatz sub-circuit, and is the most economical method available when both ansatz unitaries can be efficiently implemented.

Swap test. A more general protocol, applicable when $|\psi_i\rangle$ and $|\psi_f\rangle$ are prepared independently on two copies of the work register, is the swap test [2]. Adding one ancilla initialised in $|0\rangle$, the circuit shown in Fig. 2.5 yields an ancilla-zero probability $\text{Prob}(0_{\text{anc}}) = \frac{1}{2}(1 + |\langle \psi_f | \psi_i \rangle|^2)$, from which the overlap squared is recovered as $2\text{Prob}(0_{\text{anc}}) - 1$. The swap test is more demanding in qubits ($2n + 1$ total) and gates (n

controlled-SWAPs), but does not require explicit knowledge of either ansatz unitary, only the ability to prepare both states on demand.

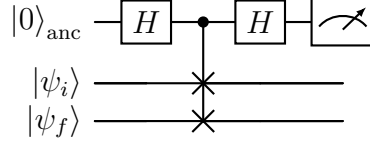


Figure 2.5: Swap test for the absolute overlap between two ansatz states $|\psi_i\rangle$ and $|\psi_f\rangle$. The Hadamard–CSWAP–Hadamard sandwich interferes the swapped and unswapped configurations of the work registers; the ancilla-zero probability is $(1 + |\langle\psi_f|\psi_i\rangle|^2)/2$. The CSWAP gate acts as a controlled exchange of the two work registers.

E1 amplitudes: matrix elements of the radial operator. For E1 transitions the relevant matrix element is the radial dipole amplitude $\langle\psi_f|r|\psi_i\rangle$, where $|\psi_i\rangle$ and $|\psi_f\rangle$ are eigenstates of the initial- and final-state channels. In the HO basis the radial operator is an $N \times N$ matrix $R_{mk} = \langle\phi_{m\ell_f}|r|\phi_{k\ell_i}\rangle$ which, like the Hamiltonian, can be evaluated in closed form. Using the same techniques that produced Eq. (2.14), one obtains, for the $P \rightarrow S$ case ($\ell_i = 1, \ell_f = 0$) and $N = 4$, the explicit matrix [12]

$$R = \begin{pmatrix} 0.5775 & -0.4715 & 0 & 0 \\ 0 & 0.8821 & -0.6668 & 0 \\ 0 & 0 & 0.8821 & -0.8167 \\ 0 & 0 & 0 & 1.0002 \end{pmatrix} \quad (\text{units of fm}). \quad (2.73)$$

The amplitude is then

$$\langle\psi_f|r|\psi_i\rangle = \mathbf{c}_f^\top R \mathbf{c}_i, \quad (2.74)$$

where $\mathbf{c}_i$ and $\mathbf{c}_f$ are the real coefficient vectors of $|\psi_i\rangle$ and $|\psi_f\rangle$ in the HO basis.

Remark (Errata). Woloshyn’s Eq. (13) reports the (2, 2) element of R as 5.4382, which is in fact the numerical value of an entry in the 1S_0 Hamiltonian matrix Eq. (2.16); the correct value, consistent with the monotonically growing radial-element pattern, is 0.8821, as used in Eq. (2.73). This is one of two copy-paste errata noted in this chapter (the other being the 1P_1 diagonal entry of Section 2.1.5).

To extract $\langle\psi_f|r|\psi_i\rangle$ on a quantum device, we first Pauli-decompose the operator R on the work register (analogously to Section 2.1.6). The matrix element then reduces to a sum of terms of the form $\langle\psi_f|P|\psi_i\rangle$ for Pauli strings P . Each such term can be estimated by the *Hadamard test* [2]: with one ancilla qubit prepared in $|0\rangle$, the circuit shown in Fig. 2.6 yields an ancilla-zero probability

$$\text{Prob}(0_{\text{anc}}) = \frac{1 + \Re \langle\psi_f|P|\psi_i\rangle}{2}, \quad (2.75)$$

from which the real part of the matrix element is obtained as $2 \text{Prob}(0_{\text{anc}}) - 1$. Replacing the leading Hadamard on the ancilla by a $S^\dagger H$ rotation yields, by the same algebra, the imaginary part of the matrix element. For real-symmetric problems such as the channel Hamiltonians considered here, only the real part is needed.

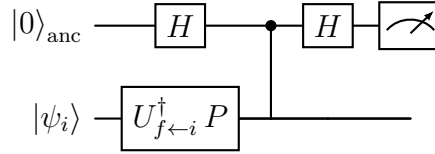


Figure 2.6: Hadamard test for the matrix element $\langle \psi_f | P | \psi_i \rangle$. Conditioned on the ancilla, the work register is rotated by an operator whose expectation value coincides with the desired matrix element. The ancilla-zero probability is $(1 + \Re \langle \psi_f | P | \psi_i \rangle)/2$. Replacing the first Hadamard by $S^\dagger H$ gives the imaginary part.

The E1 transition amplitude is then a weighted sum of Hadamard tests, one per Pauli string in the decomposition of R :

$$\langle \psi_f | R | \psi_i \rangle = \sum_k r_k \langle \psi_f | P_k | \psi_i \rangle, \quad (2.76)$$

where r_k are the Pauli coefficients of R obtained from Eq. (2.19). With this construction, the same quantum-algorithmic framework that yields energies also yields the matrix elements needed for radiative transitions, allowing a direct comparison with experimental decay widths and lifetimes once eigenstates have been obtained.

2.4 Yi et al. — Trotter–Taylor Imaginary-Time Evolution

Yi et al. [13] introduce a probabilistic imaginary-time evolution algorithm, named Trotter–Taylor Imaginary-Time Evolution (TTITE), based on combining a Trotter decomposition of the imaginary-time propagator with a truncated Taylor expansion implemented through a linear combination of unitaries (LCU). They demonstrate the algorithm on the molecular hydrogen (H_2) Hamiltonian and on one-dimensional Heisenberg spin chains. The construction itself depends only on the Pauli decomposition of the target Hamiltonian and is therefore generic; in this thesis we apply it to the channel Hamiltonians of Section 2.1.5 in the same direct encoding used by Woloshyn (Section 2.3.2), providing a third independent route to the charmonium spectrum.

2.4.1 Strategy: implement the propagator directly

In contrast to VQE and VQITE, TTITE does not fit any ansatz. The starting point is again the eigenstate-projection property of imaginary time Eq. (2.42). We seek a unitary protocol that, given the initial state $|\psi(0)\rangle$ on the work register, returns $|\psi(\tau)\rangle \propto e^{-H\tau} |\psi(0)\rangle$ to controllable accuracy. Two obstacles must be overcome.

The first obstacle is the non-unitarity of the propagator $e^{-H\tau}$. We resolve it by writing each elementary factor of a Trotter-decomposed propagator as a linear combination of two unitaries; such “LCU” operators admit a probabilistic implementation via an ancilla.

The second obstacle is that the Hamiltonian is a *sum* of Pauli strings, and exponentials of a sum cannot generally be separated. We resolve it through the Trotter formula, which decomposes $e^{-H\Delta\tau}$ into a product of exponentials of individual Pauli strings, each of which is straightforward to handle.

We now make each of these steps precise.

2.4.2 Trotter decomposition

Let $H = \sum_{i=1}^m c_i h_i$ be the Pauli decomposition of one of the channel Hamiltonians (obtained via Eq. (2.19), with the direct encoding), with $c_i \in \mathbb{R}$ and each h_i a Pauli string satisfying $h_i^2 = I$. The first-order *Trotter formula* [22, 23] approximates the imaginary-time propagator for a small step $\Delta\tau$ as

$$e^{-H\Delta\tau} = \prod_{i=1}^m e^{-c_i h_i \Delta\tau} + O((\Delta\tau)^2), \quad (2.77)$$

with leading error proportional to the commutators $[h_i, h_j]$. To evolve over a total imaginary time τ we apply $L = \tau/\Delta\tau$ such Trotter steps in succession, accumulating an overall error of order $\tau\Delta\tau$. Higher-order symmetric (Suzuki) schemes reduce this error to $O(\Delta\tau^{2k+1})$ for stage- $2k$ formulas, at the cost of more elementary exponentials per step [23]; for the small Hamiltonians of this work the first-order formula Eq. (2.77) is already accurate enough.

2.4.3 Taylor expansion of a single Trotter step

Each factor $e^{-c h \Delta\tau}$ in Eq. (2.77) is the exponential of an Hermitian operator whose square is the identity, $h^2 = I$. This property gives a remarkably simple closed form. Expanding

the exponential as a power series and separating even and odd powers,

$$\begin{aligned}
e^{-ch\Delta\tau} &= \sum_{j=0}^{\infty} \frac{(-c\Delta\tau)^j}{j!} h^j \\
&= \left(\sum_{j \text{ even}} \frac{(-c\Delta\tau)^j}{j!} \right) I + \left(\sum_{j \text{ odd}} \frac{(-c\Delta\tau)^j}{j!} \right) h = \cosh(c\Delta\tau) I - \sinh(c\Delta\tau) h,
\end{aligned} \tag{2.78}$$

$$\tag{2.79}$$

where we used $h^{2k} = I$ and $h^{2k+1} = h$. The single Trotter step is therefore exactly a linear combination of two unitaries: the identity and h itself. Truncating the series at finite order R ,

$$T^{(R)}(c\Delta\tau) \equiv \alpha I + \beta h, \quad \alpha = \sum_{\substack{j=0 \\ j \text{ even}}}^R \frac{(-c\Delta\tau)^j}{j!}, \quad \beta = \sum_{\substack{j=1 \\ j \text{ odd}}}^R \frac{(-c\Delta\tau)^j}{j!}, \tag{2.80}$$

incurs a Taylor error of order $(\Delta\tau)^{R+1}$, which for $R = 4$ or higher is negligible compared to the Trotter error $O(\Delta\tau^2)$ in Eq. (2.77). The truncation is convenient because it makes α and β polynomials in $\Delta\tau$, slightly simpler than $\cosh$ and $\sinh$.

Remark. The closed form Eq. (2.79) shows that the Taylor truncation is in fact *not necessary* for correctness; one could simply use $\alpha = \cosh(c\Delta\tau)$ and $\beta = -\sinh(c\Delta\tau)$. The finite-order form Eq. (2.80) matches the construction of Yi et al. [13] and makes the analysis of the LCU success probability slightly more uniform across different Hamiltonians, but the algorithm itself is unchanged.

2.4.4 Linear-combination-of-unitaries (LCU) implementation

The operator $\alpha I + \beta h$ in Eq. (2.80) is a non-unitary linear combination of two unitaries. Operators of this form admit a celebrated probabilistic implementation, the LCU construction [24], which uses a single ancillary qubit. We derive it explicitly here.

Augment the work register with one ancilla qubit, initially in $|0\rangle$. The combined state is $|0\rangle \otimes |\psi\rangle$. The LCU circuit applies the following three operations.

(i) Ancilla preparation. Apply $R_y(\vartheta)$ to the ancilla with

$$\vartheta = 2 \operatorname{atan2}(\beta, \alpha), \tag{2.81}$$

so that $R_y(\vartheta) |0\rangle = \cos(\vartheta/2) |0\rangle + \sin(\vartheta/2) |1\rangle = (\alpha |0\rangle + \beta |1\rangle) / \sqrt{\alpha^2 + \beta^2}$. The combined state is then

$$|\phi_1\rangle = \frac{1}{\sqrt{\alpha^2 + \beta^2}} (\alpha |0\rangle + \beta |1\rangle) \otimes |\psi\rangle. \tag{2.82}$$

(ii) Controlled application of h . Apply the controlled unitary $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes h$, which acts as the identity if the ancilla is $|0\rangle$ and as h if it is $|1\rangle$. Since h is a Pauli string of length n , the controlled gate decomposes into n elementary controlled-Pauli gates acting on the work qubits. After this step,

$$|\phi_2\rangle = \frac{1}{\sqrt{\alpha^2 + \beta^2}} (\alpha |0\rangle \otimes |\psi\rangle + \beta |1\rangle \otimes h|\psi\rangle). \quad (2.83)$$

(iii) Hadamard and measurement on the ancilla. Apply a Hadamard gate to the ancilla and then measure it in the computational basis. Using $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, the state just before measurement is

$$|\phi_3\rangle = \frac{1}{\sqrt{2(\alpha^2 + \beta^2)}} (|0\rangle \otimes (\alpha I + \beta h)|\psi\rangle + |1\rangle \otimes (\alpha I - \beta h)|\psi\rangle). \quad (2.84)$$

A measurement outcome of 0 on the ancilla projects the work register onto the (unnormalised) state $(\alpha I + \beta h)|\psi\rangle$. After renormalisation this is precisely the desired truncated Trotter step.

The probability of measuring $|0\rangle$ on the ancilla,

$$P_s = \frac{\|(\alpha I + \beta h)|\psi\rangle\|^2}{2(\alpha^2 + \beta^2)}, \quad (2.85)$$

sets the cost of the protocol: if the measurement returns $|1\rangle$ the work state has been corrupted and the segment must be restarted from $|\psi\rangle$. For small $\Delta\tau$ one has $\alpha \approx 1$ and $\beta \approx -c\Delta\tau$, so $P_s \approx 1/2$; the protocol is therefore efficient at small step sizes.

Figure 2.7 summarises the LCU circuit graphically.

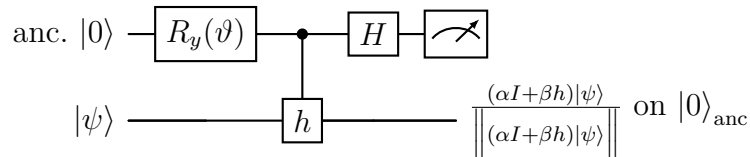


Figure 2.7: LCU circuit implementing one truncated Trotter step $T^{(R)}(c\Delta\tau) = \alpha I + \beta h$ on the work register $|\psi\rangle$. The ancilla is rotated by an angle $\vartheta = 2\arctan 2(\beta, \alpha)$, a controlled application of the Pauli string h entangles ancilla and work, a Hadamard interferes the two LCU branches, and a measurement on the ancilla post-selects the desired branch. The work register collapses to the (renormalised) truncated step exactly when the measurement outcome is 0, an event of probability P_s given by Eq. (2.85).

2.4.5 The full TTITE algorithm

Composing the building blocks of Sections 2.4.2 to 2.4.4 yields the complete TTITE procedure, summarised in Fig. 2.8.

Algorithm: TTITE.

Input: Pauli decomposition $H = \sum_{i=1}^m c_i h_i$; initial state $|\psi_0\rangle$; total time τ ; step size $\Delta\tau$; Taylor order R .

Output: an approximation to $e^{-H\tau} |\psi_0\rangle / \|e^{-H\tau} |\psi_0\rangle\|$.

1. Set $L = \lceil \tau / \Delta\tau \rceil$ and $|\psi\rangle \leftarrow |\psi_0\rangle$.
2. Repeat L times:
 - (a) For $i = 1, \dots, m$:
 - i. Compute the Taylor coefficients α, β from Eq. (2.80) with $c = c_i$ and $h = h_i$.
 - ii. Build the LCU circuit of Fig. 2.7 with angle Eq. (2.81) and controlled-Pauli string h_i .
 - iii. Execute the circuit on $|0\rangle_{\text{anc}} \otimes |\psi\rangle$; on outcome 0 update $|\psi\rangle$ with the (renormalised) post-selected state; on outcome 1 *restart* the current Trotter step.
3. Return $|\psi\rangle$.

Figure 2.8: Pseudocode for the TTITE algorithm. Each Trotter step requires m successful LCU post-selections; the per-segment success probability is given by Eq. (2.86).

The algorithm is generic in the Hamiltonian: it uses only the Pauli decomposition produced via Eq. (2.19) and the (small) number of qubits required by the encoding. In this work we apply it to the channel Hamiltonians in the direct encoding of Section 2.3.2 (two work qubits plus one ancilla, i.e. three qubits in total).

2.4.6 Success probability and resource cost

Each Trotter step requires m LCU circuits to succeed (one per Pauli term), so the segment-level success probability factorises:

$$P_{\text{seg}} = \prod_{i=1}^m P_{s,i}, \quad (2.86)$$

with $P_{s,i}$ given by Eq. (2.85). For small $\Delta\tau$ each individual factor approaches $1/2$, so the per-segment probability decays exponentially in the number of Pauli terms. Two observations soften this otherwise daunting cost.

First, Yi et al. [13] prove that the success probability is *monotonically non-decreasing* in the imaginary-time step: as the state approaches the ground state, applying $H - E_0$ becomes quasi-tangential to it, which inflates the success probabilities $P_{s,i}$. Concretely, with $\tau_\ell = \ell \Delta\tau$ denoting the cumulative imaginary time after the ℓ -th segment,

$$P_{\text{seg}}(\tau_1) \leq P_{\text{seg}}(\tau_2) \leq \dots \leq P_{\text{seg}}(\tau_L), \quad (2.87)$$

so the algorithm becomes *more* efficient as it converges.

Second, the failure probability $1 - P_{s,i}$ can be amplified toward zero by amplitude amplification, at a cost of additional ancillas and depth. We do not employ amplitude amplification in this work, instead relying on the simpler repeat-until-success strategy described in Fig. 2.8. For the two-qubit channel Hamiltonians considered here, the per-segment cost is at most a small multiplicative overhead, and the algorithm remains practical for the imaginary times required to reach the ground state.

The overall picture is therefore the following. TTITE bypasses the parameter-space optimisation of VQE and VQITE by implementing the imaginary-time propagator more or less directly on the device, at the price of an ancilla and a post-selection step per Pauli term. Its accuracy is controlled by two parameters, $\Delta\tau$ and R , and is unaffected by ansatz expressibility.

2.5 Summary

The framework developed in this chapter equips us with everything needed to reproduce the three quantum simulations of charmonium spectroscopy that this thesis benchmarks. Section 2.1 produces three 4×4 real-symmetric channel matrices for 1S_0 , 3S_1 , and 1P_1 , together with the general procedure for expanding such a matrix in Pauli strings. The Gallimore–Liao section (Section 2.2) encodes the Hamiltonian fermionically through the Jordan–Wigner transformation, uses a UCC ansatz, and minimises the energy by classical optimisation; we extend their construction to the spin-dependent $N = 4$ case for comparability. The Woloshyn section (Section 2.3) encodes the Hamiltonian directly on two qubits, uses a three-parameter R_y ansatz, evolves it under the McLachlan imaginary-time equations of motion, and computes M1 and E1 transition amplitudes. The Yi et al. section (Section 2.4) skips the ansatz entirely, implementing the imaginary-time propagator probabilistically through LCU circuits on three qubits (two work, one ancilla).

Table 2.2 collects these differences side by side. The classical and quantum workloads listed are per iteration or per Trotter step.

A few qualitative consequences of Table 2.2 deserve emphasis. First, the three papers make two distinct encoding choices: the fermionic (JW) encoding of Gallimore and Liao uses one qubit per HO basis state, whereas the direct encoding of Woloshyn and our application of Yi et al. uses one qubit per pair of basis states. The JW encoding introduces a particle-conservation constraint on the ansatz, while the direct encoding does not. Second, Gallimore–Liao and Woloshyn both optimise parameters of a small ansatz manifold, so their accuracy is bounded above by that manifold’s expressibility; Yi et al.’s scheme sidesteps the ansatz at the price of one ancilla qubit and a post-selection per Pauli term. Third, transitions (Section 2.3.9) are post-processing operations on eigenstates produced by any of the three methods; their accuracy is therefore directly inherited from the

Table 2.2: Side-by-side comparison of the three algorithms reproduced in this chapter, as applied to the channel Hamiltonians of Eqs. (2.16) to (2.18). Each method uses the encoding chosen by its source paper, with the small caveat that Gallimore and Liao’s $N = 3$ spinless setting is extended to $N = 4$ with spin in Section 2.2.7.

	Gallimore–Liao (VQE)		Woloshyn (VQITE)	Yi et al. (TTITE)
Encoding	Jordan–Wigner		Direct (compact)	Direct (compact)
Qubits (work + anc.)	$N + 0$		$\lceil \log_2 N \rceil + 0$	$\lceil \log_2 N \rceil + 1$
Pauli terms (N=4)	~ 17		~ 10	~ 10
Ansatz	UCC, $N-1$ params		R_y + CNOT, 3 params	none
What is updated	ansatz parameters		ansatz parameters	work-register state itself
What is optimised	energy expectation value		McLachlan residual norm	— (no parameters)
Classical workload	gradient-free	optimiser	small linear-system solves	track ancilla outcomes
Excited states	orthogonality constraints	constraints	state-overlap penalties	not directly supported
Dominant error	ansatz expressibility		ansatz + Trotter-like step	Trotter + Taylor + finite shots

underlying eigenstate accuracy.

The next chapter applies all three algorithms to the channel Hamiltonians of Section 2.1.5, quantifying these trade-offs through numerical experiments with the codebase developed alongside this thesis.

Chapter 3

Results

This chapter reports the numerical results obtained by applying the four computational strategies of Chapter 2 to the spin-dependent charmonium Hamiltonians of Section 2.1.5: exact diagonalisation (classical reference), VQE in the direct (2-qubit) and Jordan–Wigner (4-qubit) encodings, VQITE, and TTITE. We first run a noiseless sanity check on a simulator (Section 3.2), then re-execute the same protocols under depolarising noise to characterise the resilience of each method (Section 3.4). A brief summary (Section 3.5) closes the chapter.

3.1 Experimental setup

All quantum computations run through Qiskit’s `AerSimulator`. Pauli expectation values are estimated with $N_{\text{shots}} = 8192$ shots per term; VQITE overlaps for the metric tensor use $4 \times N_{\text{shots}}$. The classical optimiser for VQE is COBYLA with a maximum of 300 iterations. VQITE uses 40 forward-Euler imaginary-time steps with $\Delta\tau = 0.05$ and Tikhonov regularisation $\lambda = 10^{-3}$. TTITE uses total imaginary time $\tau_{\text{total}} = 5$ with step $\Delta\tau = 0.02$ and Taylor order $R = 10$. The depolarising channel of Eq. (1.13) is applied to every gate when the noisy backend is in use; rates swept are $p \in \{0.0, 0.005, 0.01, 0.02, 0.05\}$ with three independent repeats per configuration. A random seed (42) makes the runs reproducible. All energies below are reported in inverse-femtometres; multiplication by $\hbar c = 197.326 \text{ MeV}\cdot\text{fm}$ converts to MeV.

The full implementation—including the Hamiltonian builders, the three algorithm wrappers, the noise sweep driver, and the standalone visualisation script that produces every figure in this chapter—is released as open-source software at <https://github.com/Lucas-Froguel/quarkonium-simulations> [25], together with all configuration files needed to reproduce the runs reported below.

3.2 Noiseless benchmark

3.2.1 Ground-state energies

Table 3.1 reports the ground-state energy of each channel obtained by each method, alongside the exact diagonalisation reference. Figure 3.1 shows the corresponding energy convergence traces: COBYLA iterations for VQE, imaginary-time steps for VQITE, Trotter segments for TTITE.

Table 3.1: Ground-state energies in the noiseless simulator ($p = 0$). Energies in fm^{-1} ; absolute errors are $\Delta E = |E_{\text{est}} - E_{\text{exact}}|$. “VQE” uses the direct encoding (2 qubits); “VQE-JW” uses the Jordan–Wigner encoding (4 qubits, UCC ansatz). $\mathcal{F}$ is the fidelity with the exact ground state.

Channel	Method	E_{est}	E_{exact}	ΔE	$\mathcal{F}$
1S_0	VQE	0.452	0.391	0.060	0.975
	VQE-JW	0.419	0.391	0.027	0.989
	VQITE	0.376	0.391	0.016	0.99995
	TTITE	0.419	0.391	0.027	0.99984
3S_1	VQE	0.769	0.752	0.017	0.991
	VQE-JW	0.735	0.752	0.017	0.993
	VQITE	0.744	0.752	0.008	0.99990
	TTITE	0.770	0.752	0.018	0.99989
1P_1	VQE	2.780	2.781	0.001	0.998
	VQE-JW	2.745	2.781	0.036	0.999
	VQITE	7.416	2.781	4.635	0.169
	TTITE	2.785	2.781	0.003	0.99998

Three patterns emerge. First, TTITE consistently produces a high-fidelity ground state ($\mathcal{F} \gtrsim 0.9999$ on every channel), reflecting the fact that the algorithm does not depend on ansatz expressibility; its $\mathcal{O}(10^{-2}) \text{ fm}^{-1}$ energy error is dominated by the shot-noise estimator of the final expectation value rather than by any algorithmic bias. Second, the two VQE variants agree to within $\sim 10^{-2} \text{ fm}^{-1}$ across all channels, confirming that the Jordan–Wigner encoding faithfully reproduces the physics encoded by the direct mapping. Third, VQITE is the most volatile of the three: on 1S_0 and 3S_1 it recovers the true ground state with fidelity above 0.9999, but on 1P_1 the local McLachlan descent on the three-parameter manifold falls into an unphysical region (final energy 7.42, well above any low-lying eigenvalue, fidelity ~ 0.17). The failure is a known sensitivity of VQITE to the initial parameter point and to shot noise on the quantum metric estimate; VQE on the same ansatz is not similarly trapped because COBYLA samples a trust region and can escape such basins.

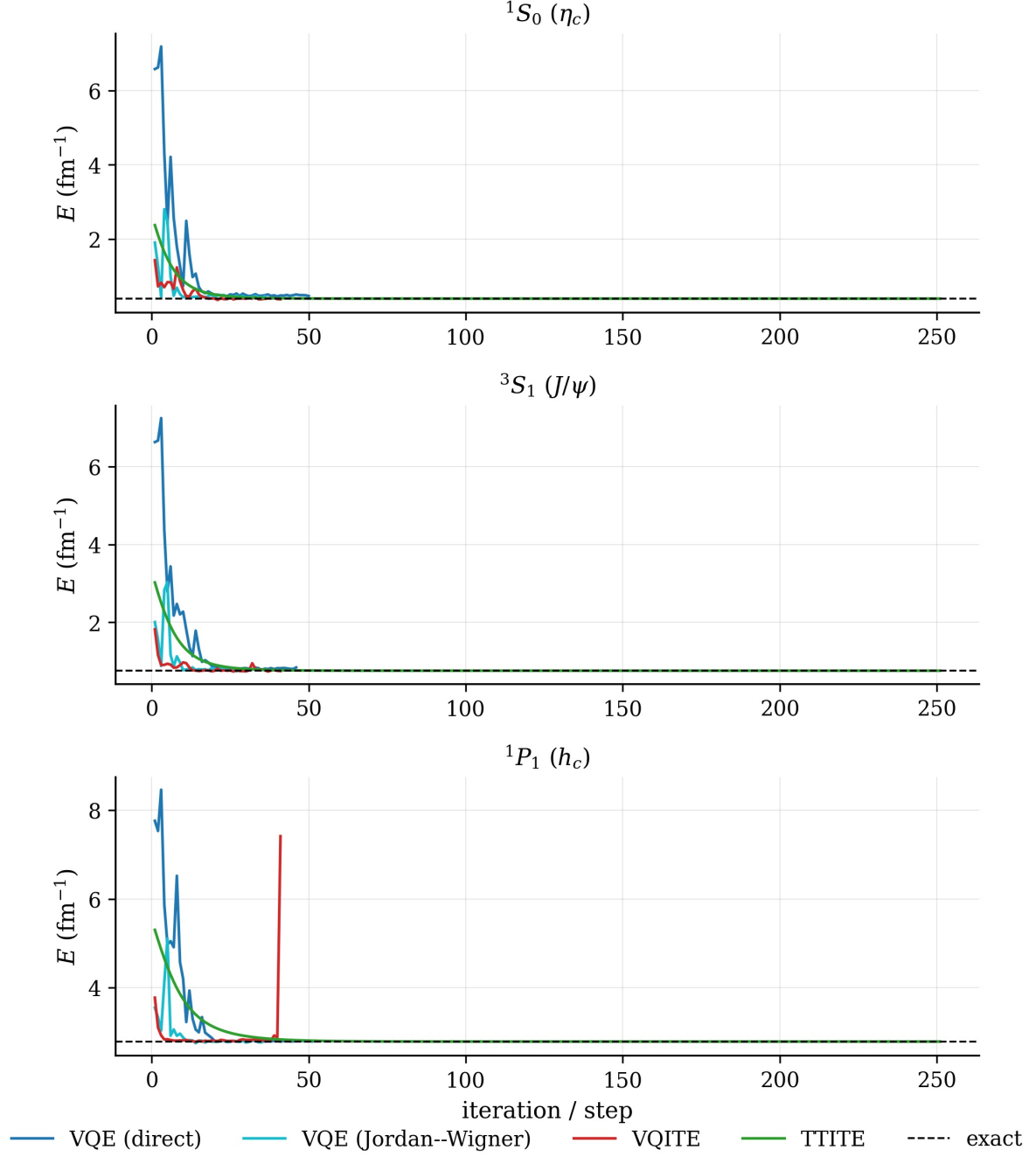


Figure 3.1: Noiseless energy convergence on each channel. The horizontal dashed line marks the exact eigenvalue. The x -axis is the COBYLA iteration for VQE/VQE-JW, the imaginary-time step for VQITE, and the Trotter segment for TTITE.

3.2.2 Ground-state error summary

Figure 3.2 collects the absolute errors of Table 3.1 into a single bar chart, separating the $\sim 10^{-3}$ accuracy of TTITE from the $\sim 10^{-2}$ ansatz-limited accuracy of the variational methods.

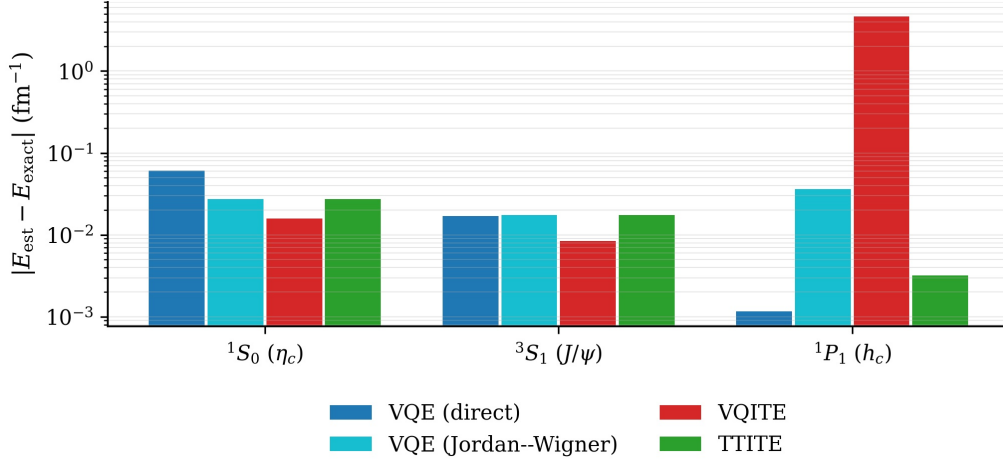


Figure 3.2: Absolute ground-state energy error per (channel, method) in the noiseless simulator (log scale). TTITE saturates the simulation precision; the variational methods are bounded above by their ansatz expressibility.

3.2.3 Computational cost

Figure 3.3 reports the wall-clock time per noiseless ground-state computation, extracted from the run logs. Three orders of magnitude separate TTITE—which on a noiseless statevector backend needs only a single chain of LCU circuits and a final shot-based energy measurement—from VQITE, whose imaginary-time loop estimates a fresh metric tensor (six overlap circuits) and a fresh parameter-shift gradient (six energy evaluations) at every step. VQE-direct and VQE-JW sit between them, with VQE-JW $\sim 20\%$ slower because of the deeper UCC circuit on twice as many qubits.

3.2.4 Physical interpretation: meson masses

The Hamiltonian of Section 2.1.5 is the internal binding Hamiltonian of the charm–anticharm pair, so the predicted meson mass is obtained by adding back the rest energy of the two constituent quarks,

$$M_{\text{meson}} = 2m_c + E_0 \cdot \hbar c, \quad (3.1)$$

with $m_c = 1.4794$ GeV and $\hbar c = 197.326$ MeV·fm. Applying Eq. (3.1) to the exact-diagonalisation eigenvalues yields the predictions of Table 3.2, which we compare with

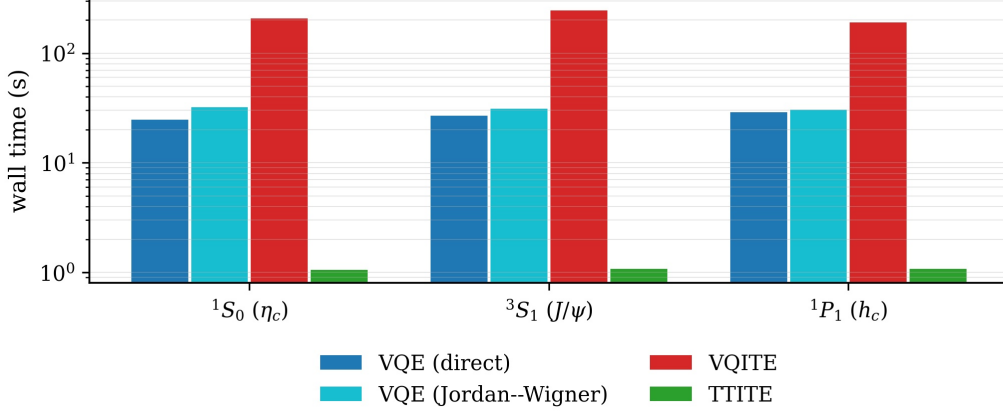


Figure 3.3: Wall-clock time of a single noiseless ground-state run on each channel (log scale). VQITE pays for its quantum metric tensor by $\mathcal{O}(50\times)$ in wall time relative to VQE; TTITE is two orders of magnitude faster than either variational method on the noiseless statevector backend.

the experimentally measured masses of the corresponding mesons from the Particle Data Group [26].

Table 3.2: Predicted ground-state meson masses obtained from the exact diagonalisation of Eqs. (2.16) to (2.18) via Eq. (3.1), compared with experimental values for the corresponding mesons [26]. All masses in MeV.

Channel	Meson	E_0 (fm $^{-1}$)	M_{pred}	M_{exp}	ΔM
1S_0	η_c	0.391	3036	2984	+52
3S_1	J/ψ	0.752	3107	3097	+10
1P_1	h_c	2.781	3508	3525	-17

The agreement with experiment is at the 10–50 MeV level, typical for a four-state harmonic-oscillator truncation of the Cornell model. A particularly stringent test is the J/ψ – η_c hyperfine splitting, where the difference of the two predictions cleanly cancels the rest-mass offset:

$$\begin{aligned}\Delta M_{\text{hf}}^{\text{pred}} &= (E_0^{^3S_1} - E_0^{^1S_0}) \hbar c = 71.2 \text{ MeV}, \\ \Delta M_{\text{hf}}^{\text{exp}} &= M_{J/\psi} - M_{\eta_c} = 113 \text{ MeV}.\end{aligned}$$

The model under-predicts the splitting by roughly 40 MeV, a known artefact of using a Gaussian-regularised contact term without the full spin–orbit corrections of Eq. (2.3).

3.2.5 Excited states

We compute the four lowest eigenvalues per channel using the orthogonalisation/penalty strategy of Section 2.2.6. Both VQE encodings are run; the VQITE column of Table 3.3

is reserved for follow-up work—the same penalty machinery (Eq. (2.69)) applies, but the per-step circuit overhead makes a full sweep impractical within the time budget of this thesis. TTITE does not support excited states directly and is omitted from this table.

Table 3.3: Ground and excited-state energies (fm^{-1}) obtained by penalty-based VQE. E_{exact} is the eigenvalue of the classical 4×4 diagonalisation. “—” entries are reserved for VQITE, which is left for future work.

Channel	k	E_{exact}	VQE	VQE-JW	VQITE
1S_0	0	0.391	0.429	0.472	—
	1	3.483	3.489	3.466	—
	2	5.643	5.573	5.519	—
	3	7.545	7.048	7.509	—
3S_1	0	0.752	0.808	0.776	—
	1	3.633	3.656	3.678	—
	2	5.727	7.458	5.927	—
	3	7.646	5.660	6.179	—
1P_1	0	2.781	2.798	2.772	—
	1	4.883	4.889	4.923	—
	2	6.756	7.882	6.779	—
	3	8.770	7.058	5.613	—

Figure 3.4 shows the same information graphically. The ground and first excited states are recovered with errors of at most $\sim 0.08 \text{ fm}^{-1}$ across all channels and both encodings. Higher states ($k = 2, 3$) are more difficult: the penalty cost in Eq. (2.69) stiffens the optimisation landscape as the already-found subspace grows, and gradient-free COBYLA can converge to the wrong eigenstate of the constrained problem. In several 3S_1 and 1P_1 entries the penalty method exchanges the identities of $k = 2$ and $k = 3$ (e.g. VQE on 3S_1 assigns $E_2 \approx 7.5$, near the exact $E_3 = 7.65$, and $E_3 \approx 5.7$, near the exact $E_2 = 5.73$). VQE-JW is noticeably more reliable on the third level for 1S_0 and 1P_1 , though it inherits the same $k = 3$ degradation elsewhere.

3.3 Transition amplitudes

The machinery of Section 2.3.9 converts the ground-state wavefunctions computed above into transition matrix elements. We report two cases:

- the M1 amplitude $|\langle \psi(^1S_0) | \psi(^3S_1) \rangle|^2$ for $J/\psi \rightarrow \eta_c \gamma$, evaluated as the squared overlap of the two singlet/triplet S -wave ground states (Eq. (2.71));
- the E1 amplitude $\langle \psi(^1S_0) | r | \psi(^1P_1) \rangle$ for $h_c \rightarrow \eta_c \gamma$, evaluated via the radial-operator matrix Eq. (2.73) and the inner product Eq. (2.74).

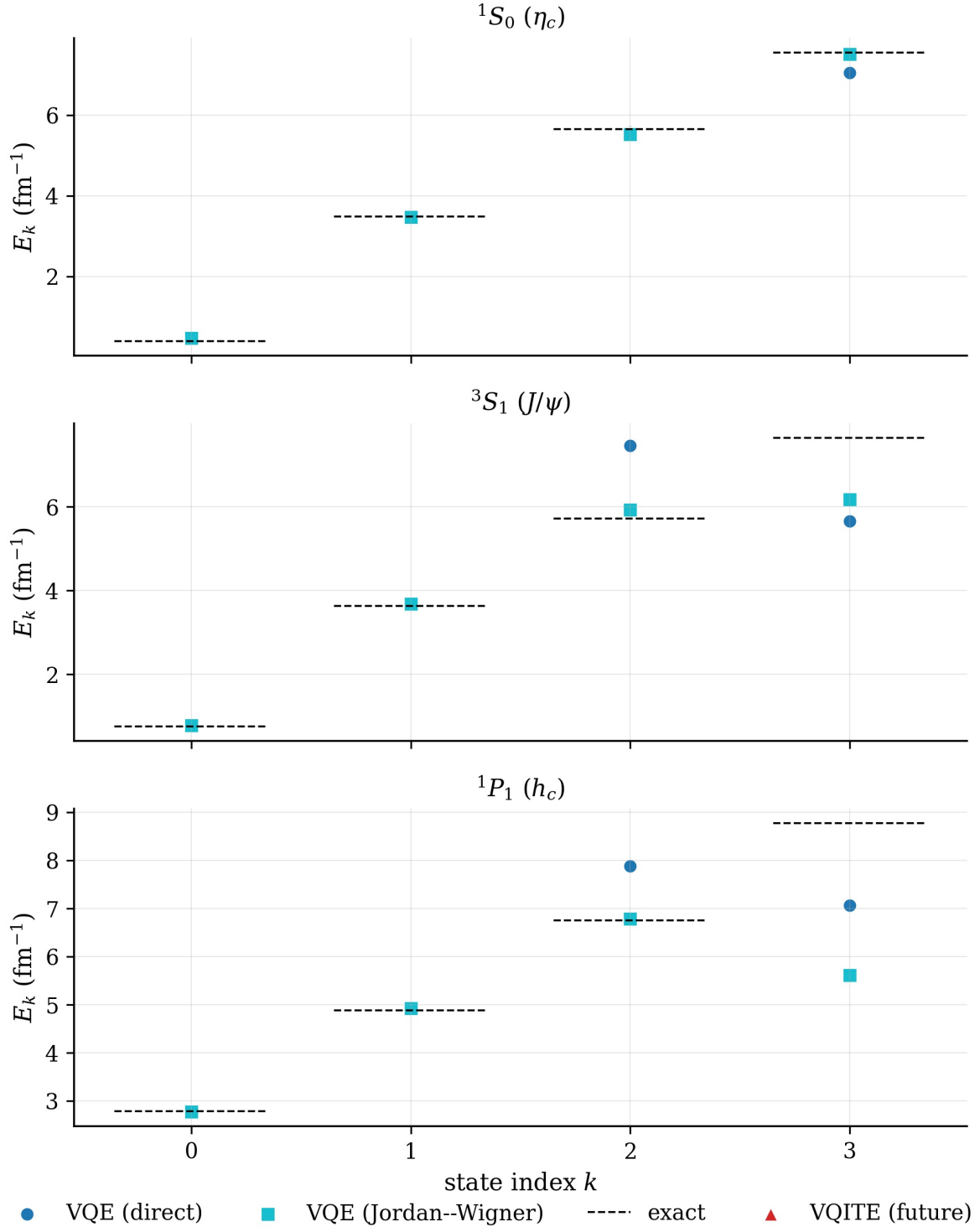


Figure 3.4: Energy-level diagram for the three channels. Black dashes mark the exact eigenvalues; coloured markers are the penalty-based VQE estimates in both encodings. The slot for VQITE excited states is preserved for follow-up work.

For VQE-JW we extract the four physical amplitudes from the 16-dimensional Jordan–Wigner statevector by projection onto the single-particle sector, ensuring an apples-to-apples comparison with the direct-encoded methods. Table 3.4 collects the results. For E1 we report $|\langle\psi^{(^1S_0)}|r|\psi^{(^1P_1)}\rangle|$ because the sign of the variational state is gauge-dependent (a converged optimisation may land on $\pm|\psi\rangle$).

Table 3.4: Transition amplitudes computed from the noiseless wavefunctions of Table 3.1. The classical reference is obtained from the exact diagonalisation of Eqs. (2.16) to (2.18). The E1 matrix uses the corrected r -matrix of Eq. (2.73).

Method	$ \langle\eta_c J/\psi\rangle ^2$ (M1)	$ \langle\eta_c r h_c\rangle $ (E1, fm)
exact	0.993	0.498
VQE	0.986	0.505
VQE-JW	0.998	0.498
VQITE	0.992	0.165
TTITE	0.993	0.496

Both M1 amplitudes are within 1% of the classical reference for every method, including the methods whose ground-state energy estimators are noticeably biased: the overlap is insensitive to the specific energy estimator and only requires the wavefunction to be close to the exact eigenstate. The E1 amplitudes are more demanding. VQE-JW and TTITE agree with the exact reference to better than 1%. The direct-encoding VQE gets the magnitude right (0.505 vs. 0.498 fm); the value of 0.165 fm returned by VQITE is the direct consequence of its catastrophic 1P_1 ground-state failure (Section 3.2.1). Together with the energy and fidelity data, this confirms that the radiative observables of the charmonium spectrum are recovered cleanly whenever the underlying state is.

3.4 Noisy benchmark

We now switch the simulator backend to `AerSimulator(noise_model=make_noise_model(p))` and re-run the ground-state computation for each method at the five depolarising rates listed in Section 3.1. Figure 3.5 plots the absolute energy error against p for the three channels; Fig. 3.6 plots the fidelity. Both VQE variants are included so that the encoding’s effect on noise resilience is visible directly.

The four methods degrade in qualitatively different ways. Among the variational algorithms, the direct-encoding VQE is the most resilient: its shallow R_y –CNOT ansatz (Fig. 2.3) is exposed to only five gates per Pauli expectation, and the gradient-free optimiser absorbs the residual shot+gate noise into the search. The Jordan–Wigner VQE, despite using the same VQE skeleton, degrades much faster: the UCC ansatz of Fig. 2.1 chains an X gate, six CNOTs and three controlled- R_y rotations on twice as many qubits,

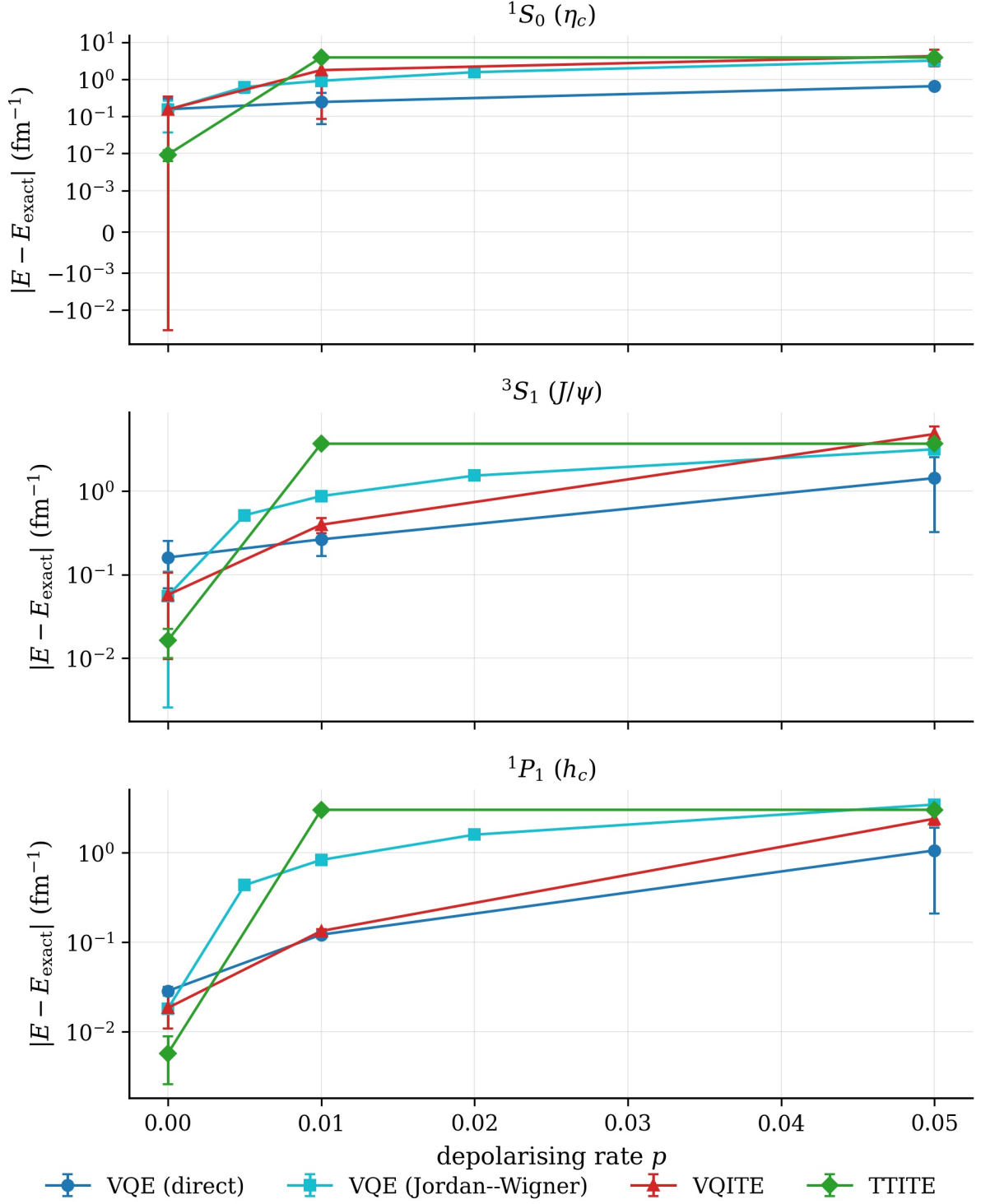


Figure 3.5: Absolute energy error vs. depolarising rate p for each channel. Error bars are the standard deviation over three repeats. The y -axis is symmetric-log to capture both the sub- 10^{-3} noiseless regime and the $\mathcal{O}(1)$ degradation at $p = 0.05$.

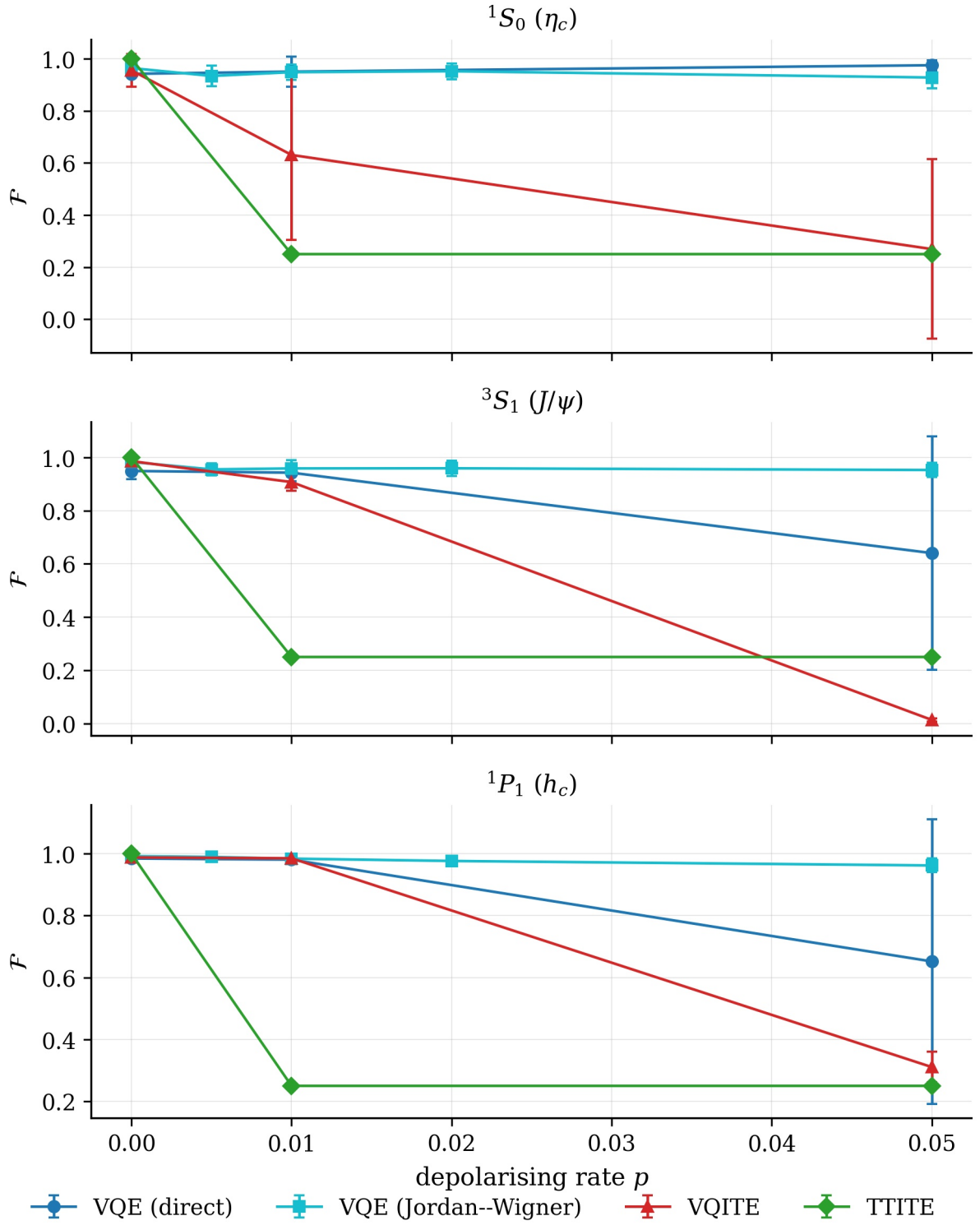


Figure 3.6: Ground-state fidelity vs. depolarising rate p for each channel. Error bars are the standard deviation over three repeats.

so the noisy Pauli estimator at the optimum is biased by an amount that grows roughly linearly with p . At $p = 0.05$ the energy error of VQE-JW is $\sim 3 \text{ fm}^{-1}$ on every channel, whereas VQE-direct stays below $\sim 1.5 \text{ fm}^{-1}$. The encoding choice that was harmless in the noiseless regime thus becomes the dominant factor on noisy hardware, and confirms a recurring lesson of NISQ-era variational algorithms: circuit depth is the noise dial.

VQITE degrades faster still, because the metric tensor and parameter-shift estimates of Eqs. (2.64) and (2.65) both require differences of noisy expectation values, so the effective signal-to-noise ratio of every step drops more sharply with p than for a direct energy minimisation; the fidelity drops below 0.3 by $p = 0.05$.

TTITE collapses outright. With $\sim 2,250$ sequential noisy gates over a full Trotter trajectory and an ancilla-reset protocol that *cannot* reject failed LCU branches on real hardware, the work register relaxes toward the maximally mixed state $\text{Tr}(H)/4$, and the fidelity drops to ~ 0.25 (random guessing in the four dimensional Hilbert space) at any non-zero noise rate. The two imaginary-time methods that were superior in the noiseless regime are therefore the most fragile under depolarising noise.

3.5 Summary

In the noiseless regime all four methods recover the channel eigenvalues, but they are not interchangeable: TTITE is essentially exact (fidelity above 0.9999 everywhere, energy error limited only by shot noise in the final Pauli measurement), VQE in either encoding is limited by the chosen ansatz at the $\sim 10^{-2} \text{ fm}^{-1}$ level, and VQITE recovers the ground state on 1S_0 and 3S_1 but falls into a local-descent trap on 1P_1 . The transition amplitudes $|\langle \eta_c | J/\psi \rangle|^2$ and $|\langle \eta_c | r | h_c \rangle|$ are recovered to within 1% of the classical reference for every method whose ground state is correct, confirming that the radiative observables of charmonium follow the same accuracy ranking as the energies.

Under depolarising noise the ranking inverts and sharpens: the shallow-circuit VQE-direct is the most resilient, the deeper-circuit VQE-JW degrades to errors of $\mathcal{O}(\text{fm}^{-1})$ at moderate p , VQITE degrades further still, and TTITE collapses to the maximally mixed state at any non-zero noise rate. The encoding choice that was second-order in the noiseless regime becomes the dominant factor in the noisy one. These observations motivate the noise-mitigation strategies developed in the next chapter.

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